

Development of a Web-based Question Bank for Automated Question Paper Generation

By

Md. Monir Ahamed

POST GRADUATE DIPLOMA
IN
INFORMATION AND COMMUNICATION TECHNOLOGY

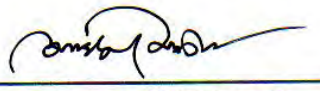


Institute of Information and Communication Technology
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

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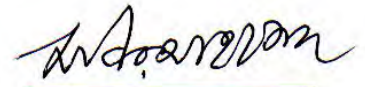
The project titled “**Development of a Web-based Question Bank for Automated Question Paper Generation**” submitted by **Md. Monir Ahamed**, Student ID.: 0412311016, Session: April, 2012, has been accepted as satisfactory in partial fulfillment of the requirement for the degree of Post Graduate Diploma in Information and Communication Technology on 9th March, 2019.

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A handwritten signature in black ink, appearing to read 'Md. Monir Ahamed', written over a horizontal line.

Md. Monir Ahamed

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List of Abbreviations

Terms	Abbreviations
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
JSP	Java Server Pages
MVC	Model View Controller
MySQL	My Structured Query Language
SDLC	System Development Life Cycle
OTP	One Time Password
JSC	Junior School Certificate
SSC	Secondary School Certificate
HSC	Higher Secondary Certificate
LAMP	Linux, Apache, MySQL, PHP/Python
SQL	Structured Query Language
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
qBank	Question Bank
QGS	Question Generation System
ERD	Entity Relation Diagram
GUI	Graphical User Interface
UI	User Interface
UML	Unified Modeling Language
URL	Uniform Resource Locator
WAF	Web Application Framework
WWW	World Wide Web
XAMPP	X (cross platform), Apache, MySQL, PHP, Perl
XHTML	Extensible Hypertext Markup Language
WYSIWYG	What You See Is What You Get

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Abstract

Question paper is an important part of an education system for assessing the learning of the students. Security is an important concern in the question paper generation process for examinations. Question paper leaks can ruin the whole examination process. This project introduced a web-based secure automated question paper generation system that is robust to question paper leakage as it replaces the manual process of question paper generation in a traditional way. The proposed system presents a smart question paper generator to generate questions within seconds. This project includes two processes- (i) Question Bank Development process and (ii) The Question Paper Generation process. Only registered composer, i.e., teacher will have the authority of addition and update in Question Bank Database. This system considered the two kinds of major examination held in our country. One is Question Paper Generation for public examination, i.e., JSC, SSC, HSC, etc., and second is Question Paper Generation for another day to day examination like a class test, term finals, and yearly test. For public examination, a registered moderator only can set the pattern, i.e., the total number of questions, chapter and complexity level of each question. But, the person does not know particularly which questions are being generated. Here, authorized exam center can print question paper after the stipulated time. Whenever any authorized exam center requests to generate question paper, a random one time password (OTP) will be generated after every successive generation of the question paper to ensure the security of the question which will be sent to a specified cell phone or email address. This OTP is used to download or print the question paper from the secure server.

For other examinations, authorities of registered institutes will be able to log in the system and customize the questions as per syllabus, requirements the complexity of the generated question paper. This project assumes that the complexity level and selecting the number of question are the main challenges in question paper generation. Other challenges include the addition of the Bangla and English version of the questions simultaneously. Hence, considering these challenges, an automated question paper generation system will be able to provide a secure environment and reduce human effort by maintaining a fair examination.

CHAPTER 1

INTRODUCTION

1.1 Background and Problem Statement

Examinations are an essential part of different stages of education and their results are used for several purposes like improvement in learning, grading and classification of students, selection for admission to higher classes or for scholarship, certification and for providing guidance for future, etc. In Bangladesh, written examinations are the most commonly used to determine the performance of a student [1]. Hence, question paper plays a vital role in the overall examination process. Presently, in most of the educational institutions question paper are set manually. For a long time, the government press printed the papers and then that was distributed to authorities in the zonal office at remote areas several days or months before the test [2]. Usually, sealed envelopes containing the question papers are used to distribute question papers from the central printing press to Zonal Offices (ZOs) and from ZOs to the local examination centers [3]. This traditional method of preparing question paper is very challenging, tedious and time consuming for the instructors [4]. During this system of reaching the question papers to their final destination, there are a large number of people involved right from those in the printing press, those in logistics of the delivery, teachers, officials and employees in places where the papers are stored and also political goons. The risk of question paper leakage compounds with every step of the process, where question paper leaks during public examinations can be prevented with the use of technology. This study proposed an integrated automated system [5]. Which automatically generates question paper from an existing question bank with the help of only authorized users, which are then printed in different locations hours before the examination. The system also offers a security function, known as one-time password (OTP), to print or download the question paper. This system uses not an only public examination but also it is very useful for various educational institutes to generate class test, terminal test, and yearly test. Here, they also have their own choice to customize the question paper

formats [6]. This system provides is suitable for all secondary and higher secondary level educational institutions in Bangladesh.

1.2 Objective with specific aims and possible outcome

The main objective of this project is to reduce the manual work load in question paper generation.

(a) Objectives with specific aims:

- To propose a framework for automated question paper repository for examiners.
- To develop a software application using web technology for automated question generation where authentication, confidentiality and integrity of the process will be maintained.
- To implement a two way security technique for downloading the question paper.

(b) Possible outcome:

The possible outcome of this project will be a software application which will be used by educational institutions for developing and maintaining a secure question repository. The system will help to automate the process of question paper generation at lesser manpower, time, energy and money.

1.3 Organization of the Report

This report is organized into five chapters.

- ❖ Chapter 1 presents background of question paper generation system development difficulties, objective of the project and layout of this report.
- ❖ Chapter 2 presents system study and requirement analysis.
- ❖ Chapter 3 presents the implementation of question paper generation system.
- ❖ Chapter 4 presents outcome of the proposed question paper generation system.
- ❖ Chapter 5 presents the conclusion and scope of the future work.

This report ends with an appendix containing the source code of project.

CHAPTER 2

SYSTEM STUDY AND REQUIREMENT ANALYSIS

2.1 The Concept of Web-based Question Bank

A web-based question bank is a large systematic collection of questions stored in the database in the order of separate unit and subject [7]. The administration is responsible for registering the users to access the question bank. The registered staff, i.e., composer of questions will have the authority of adding the questions in the database by selecting the subject, group, chapter, difficulty level, etc. A registered moderator will review the questions and then approve to save in the Question bank. The person also can perform other activities like view/update. Security of the overall system will be maintained through password protection.

2.2 Analysis of the main feature

This project is aimed to develop a web-based question bank for automated question paper generation. This chapter discusses about all functional and non-functional requirements of the proposed system. The main features of automated question paper generation are described in the following sections.

2.2.1 Automated Question Paper Generation

In the public examination (i.e., JSC, SSC, and HSC) process, a moderator only can set the pattern, i.e., total number of question per chapter and complexity level of each question. Whenever the moderator requests to generate question paper, the system extracts all the questions of the respective subject and unit. It then randomly selects the required number of questions [8]. For security reasons, the moderator remains unaware of which questions were finally generated. An authorized exam center can

print question paper after the stipulated time. For other examinations like the class test, terminal test, and yearly test, the authorized institute can log in the system and generate a question paper with own syllabus and customized formats.

2.2.1.1 System Actors

The actor is basically an impetus of any system. The control system is handled by the actor and it plays an important role in allocating a sub-division of role and control over its secondary actors. The proposed system has four types of actors that play the role of operating the system. They are listed below in (Figure 2.1)-

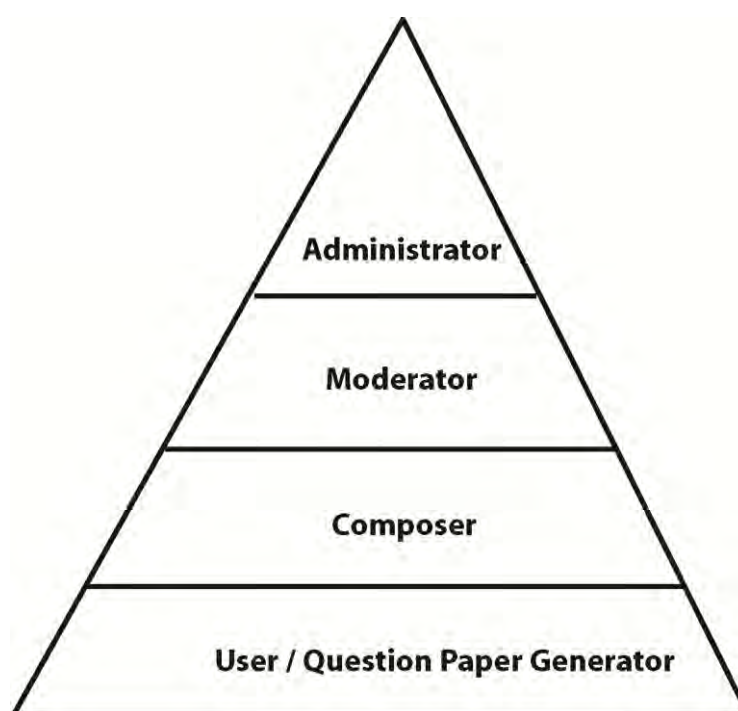


Figure 2.1: Actor hierarchy of question bank

(a) Administrator

- User(i.e., Question Generator, Composer and Moderator) Management
 - Add user
 - Edit user
 - Update user
 - Delete user

- Reset password
- Question Bank Management
 - Create subject and syllabus design
- Question Paper Management
 - Select the moderator for question paper set
 - Manage exam center
 - Approved the question print request

(b) Moderator

- Question Bank Management
 - Manage chapter and group/unit of each subject
 - Set number of question per chapter per subject
 - Activate/Deactivate questions from Question Bank, Set Marks Distribution, Disable or Delete question from the Question Bank
 - Review and final correction of the question
- Question Paper Moderation
 - Set question pattern, i.e., total number of question, question type etc.
 - Set complexity level, i.e., easy, medium, and difficult.

(c) Composer

- Create new question
- View question

(d) Exam Center/User

- Print or download question paper

2.2.1.2 User Registration

Using this Application module, system administrator can add registration related information of the user (i.e., Composer, Moderator and Question Paper Generator) to the system and can provide a user Id and password.

2.2.1.3 Subject and Syllabus Module

Using this module, system administrator can specify the subject code, and the moderator can specify group code, chapter code, number of questions of each chapter and group/unit.

2.2.1.4 Question Pattern Design Module

With the help of this module, moderator can specify the question type, complexity level, marks and question formats of each subject.

2.2.1.5 Question Aggregator Module

Using this module, Composer can enter any numbers of questions in the question bank with unique key such as question id, subject id, chapter id and group/unit id.

2.2.1.6 Question Paper Generator Module

Using this module, user can arbitrarily generate question paper according to specified request. It is the main function of the system.

2.2.2 Required Technologies and Tools

This section describes the necessary software programs and technologies and generally used to develop such application. All programs discussed here were the most recent versions at the time of working the project.

2.2.2.1 PHP

PHP is a server-side scripting language originally designed for web development to produce dynamic web pages but also used as a general-purpose programming language. It was originally created by Rasmus Lerdorf in 1994, the PHP reference implementation is now produced by The PHP Group [9]. For this purpose, PHP code is embedded into the HTML source document and interpreted by a web server with a PHP processor module, which generates the web page document. Because PHP code is executed server-side, the client will never see the code generating part or all of the

HTML content of a web page. PHP can be used in combination with various web template systems, web content management systems, and web frameworks. PHP code is usually processed by a PHP interpreter implemented as a module in the web server or as a Common Gateway Interface (CGI) executable [10]. The web server combines the results of the interpreted and executed PHP code, which may be any type of data, including images, with the generated web page.

2.2.2.2 MySQL

MySQL is a freely available open source Relational Database Management System (RDBMS) that uses Structured Query Language (SQL) [11]. MySQL is very popular for web-hosting applications because of its plethora of web-optimized features like HTML data type. It is a central component of the LAMP (Linux, Apache, MySQL, PHP/Python), open-source web application software stack that is frequently used to deliver and support advanced Web applications [12]. MySQL runs the back-end databases of some famous websites, including Wikipedia, Google, Facebook, and Twitter [13][14][15].

2.2.2.3 Java Script

JavaScript is a scripting or programming language for web. It is widely used as part of web browsers, which allow client-side scripts to interact with the user and control the browser. It communicates asynchronously and alters the content of the document that is displayed. It is embedded to the HTML page by the following tag:

```
<script type="text/javascript"> Code </script>
```

2.2.2.4 HTML5

HTML5 is a Hypertext markup language used for structuring and presenting content on the World Wide Web. It is the fifth and current major version of the HTML standard [16]. HTML is written in the form of HTML elements consisting of tags, enclosed in angle brackets (like <html>), within the web page content. HTML tags most commonly come in pairs like <h1> and </h1>, although some tags, known as empty elements, are unpaired, for example . The first tag in a pair is the start tag, the second tag is the end tag (they are also called opening tags and closing tags).

HTML5 includes detailed processing models to encourage more interoperable implementations; it extends, improves and rationalizes the markup available for documents, and introduces markup and application programming interfaces (APIs) for complex web applications. The nested nature of HTML can be seen in its document object model (DOM) (Figure 2.2)[17].

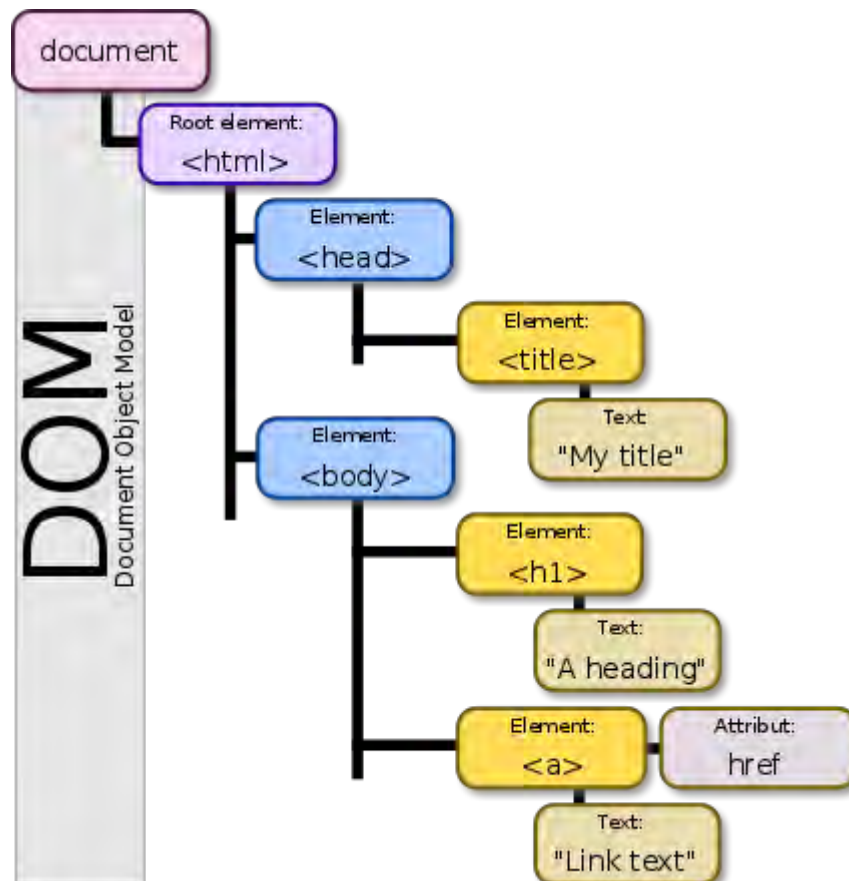


Figure 2.2: Example of DOM hierarchy in an HTML document [18]

2.2.2.5 Codeigniter Framework

This project uses CodeIgniter as the framework to demonstrate the building of the web application mentioned in its statement. CodeIgniter is a web application framework that is open source used to build dynamic PHP applications. It is based on the Model-View-Controller (MVC) development pattern. For this reason, the structures of the generated code are more structured and have clear standards. It is produced very friendly URLs. Code Igniter minimized use of \$_GET and replaced with a URL. The working principal of CodeIgniter is shown in Figure 2.3

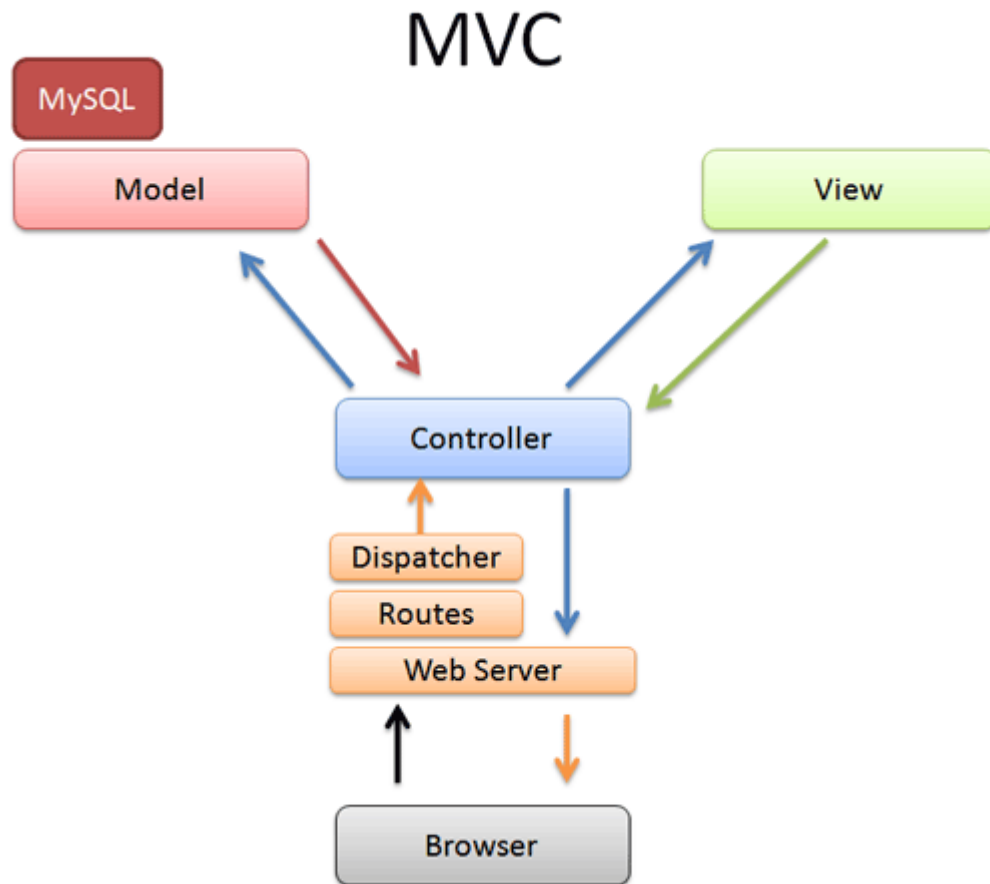


Figure 2.3: Structural Work CodeIgniter [19]

The structure of CodeIgniter starts from a browser which will interact via Controller. Then, the controller will receive and reply to all requests from the browser. For data, the controller will ask for the model and for the template, the controller will ask to view. When a browser requests a web page, then the router will find the controller which should handle the request; later the controller will use to access the data and the model view to display the data.

2.2.2.6 CkEditor 4

CKEditor is an online WYSIWYG (What You See Is What You Get) editor that is used to edit HTML documents in the browser. The benefit of CKEditor is that the user does not need to see the HTML code directly nor understand its intricacies. The editor acts as an intermediary — it hides the HTML code from the coder and lets the user work in WYSIWYG way. CKEditor 4 has features simulator to word processors

such as styles formatting (bold, italic, underline, bulleted and numbered lists), tables, web resource linking, safe undo function, image inserting, copy and paste from Word and other common HTML formatting tools [20]. This project incorporates CKEditor for image and equation inserting, text formatting and conversion of MS Word contents to HTML code.

2.2.2.7 XAMPP

XAMPP is a web server application stack. The major components it includes are Apache HTTP server, MySQL database server, Perl and PHP scripting language modules for Apache HTTP. It is basically a local host or local server. This local server works on any desktop or laptop computer. XAMPP Control Panel Application included in the release allows for an easy administration of the various server software packages. The control panel application of XAMPP is as shown in Figure 2.4.

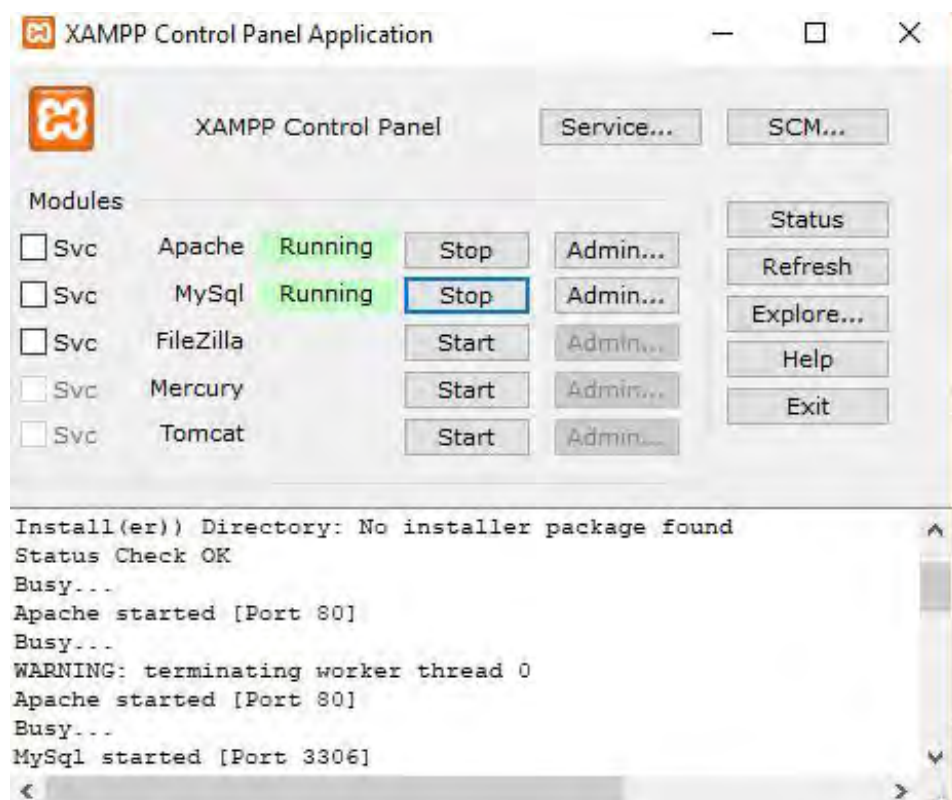


Figure 2.4: XAMPP Control Panel Application

2.2.2.8 NetBeans

NetBeans is an open-source integrated development environment (IDE) for developing applications based on Java, PHP, C++, and other programming languages. NetBeans is also referred to as a platform of modular components used for developing Java based desktop applications [21]. It offers support for third-party plug-ins with which its capabilities can be extended when needed.

2.3 System Architecture

2.3.1 Web Application

A web application is an application program that runs on a web server. Most of the desktop applications today are migrated to web applications because web applications offer a simple, intuitive and responsive user interface that let developers' relaxation on platform constraints and installation requirements. Web apps must be accessed through a web browser. Web apps have several advantages over desktop applications. Since they run inside web browsers, developers do not need to develop web apps for multiple platforms. For example, a single application that runs in Chrome will work on both Windows and OS X. Developer does not need to distribute software updates to users when the web apps are updated. Once the application on the server is updated, all users will have access to the updated version.

2.3.2 Web Server

A web server is a system that delivers content or services to end users over the internet. It is a place where all the data of a website can be saved. Every web server has an IP address and a name. When a search address is entered through the web browser, it is converted into an internet protocol where the server is located. It sends a request to the server which gets the required page and sends it to the browser. Any device can be converted into a server with the help of specific software and the availability of the internet.

2.3.3 Application Server

An application server is a software framework that provides data and code integrity by allowing more centralized approach to updates and upgrades to applications. They provide security by centralizing the management of data access and the authentication process. An application server can be used to develop and run web-based applications. There are many different types of application servers, including Java, PHP, and .NET Framework application servers.

2.3.4 The Client Site

A client (i.e., the computer, laptop, mobile, etc.), which requests the resources, through the internet with a User Interface (typically a web browser) for presentation purposes.

2.3.5 The Admin Site

Admin also another client (i.e., the computer, laptop, mobile, etc.), which requests the resources, through the internet with a user interface (typically a web browser) for creating, updating and deleting information.

CHAPTER 3

SYSTEM DESIGN AND DEVELOPMENT

3.1 Introduction

The traditional system development life cycle (SDLC) was used to develop the system. SDLC system is an approach of generating a static structure, imitation and mechanism which be used by ordinary users to explore a system. It helps to demonstrate the overall view of different phase of development procedure. Figure 3.1 shows SDLC Phases.

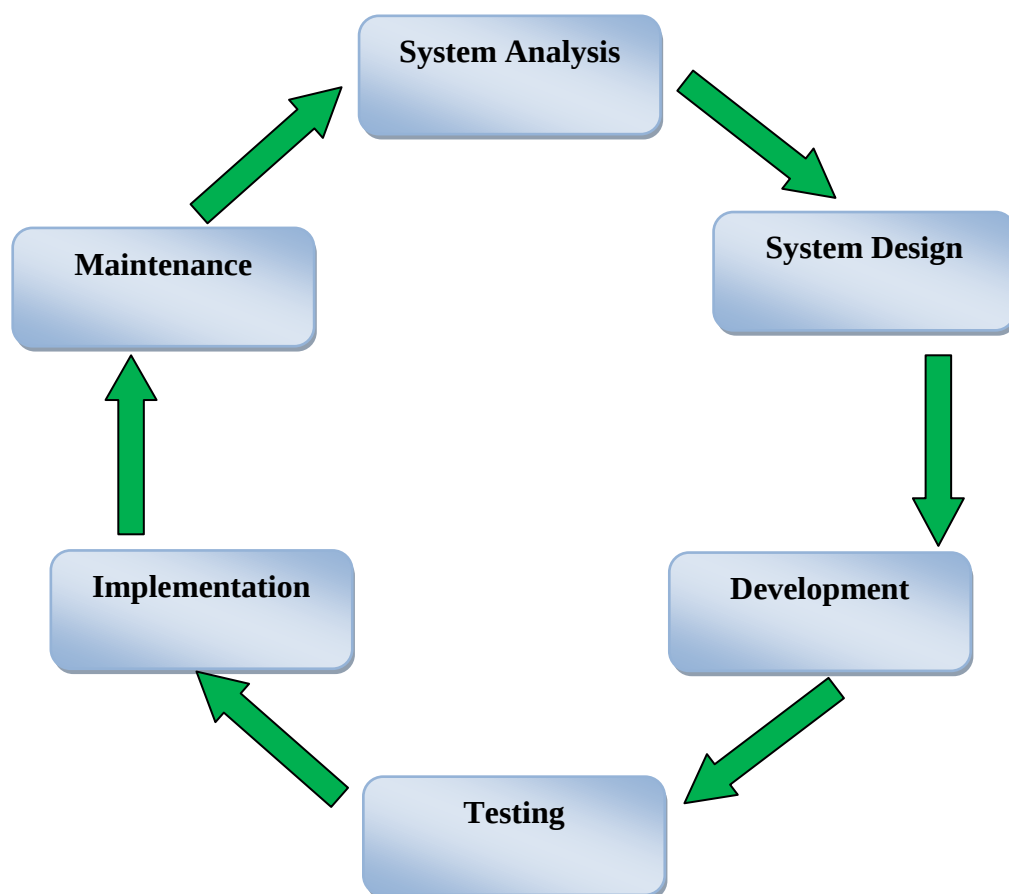


Figure 3.1: System Development Life Cycle (SDLC)

3.2 Modeling System Requirement

After performing system study and requirement modeling, a tentative framework of Question Paper Generation System was developed. A use-case diagram, entity-relationship diagram, and context diagram were developed for further processing. The design of a data flow diagram with suitable input, process, storage, and output was developed at this stage.

3.2.1 Use case Diagram

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. It is a set of scenarios that describes an interaction between a user and a system. The two main components of a use case diagram are use cases and actors. It can be shown by the Figure 3.2

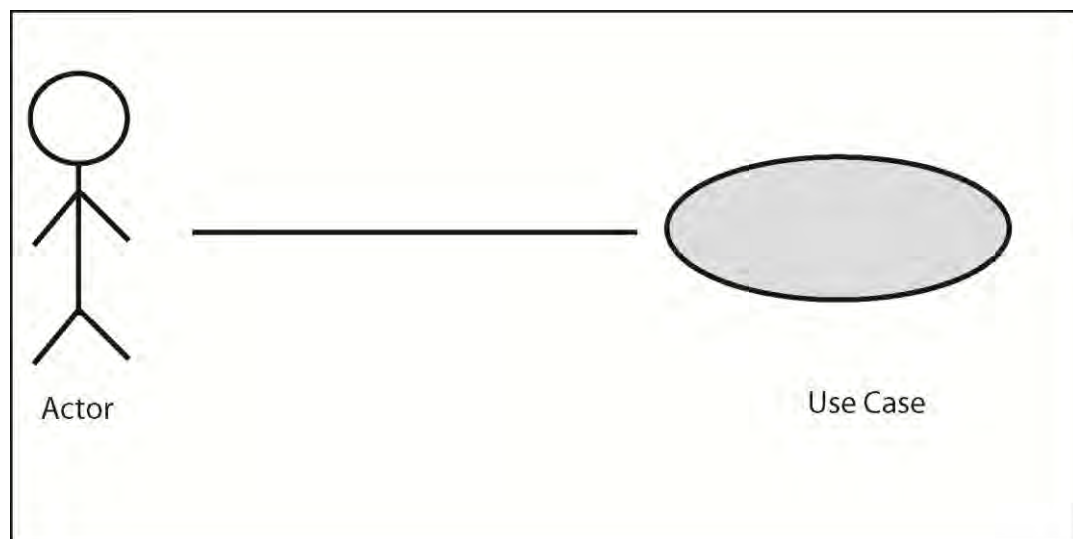


Figure 3.2: Actor and Use Case.

An actor represents a person, organization, or external system that will interact with this system. The symbols of actors are drawn as stick figures. A use case is an external view of the system that represents some actions the user might perform in order to complete a task and is drawn as a horizontal ellipse. Lines are used to representing the relationships between these elements.

3.2.1.1 Use Case Diagram of Administrator

Figure 3.3 shows the use case diagram of an administrator. The administrator can log in his/her account and can insert, update and delete the required information from the system.

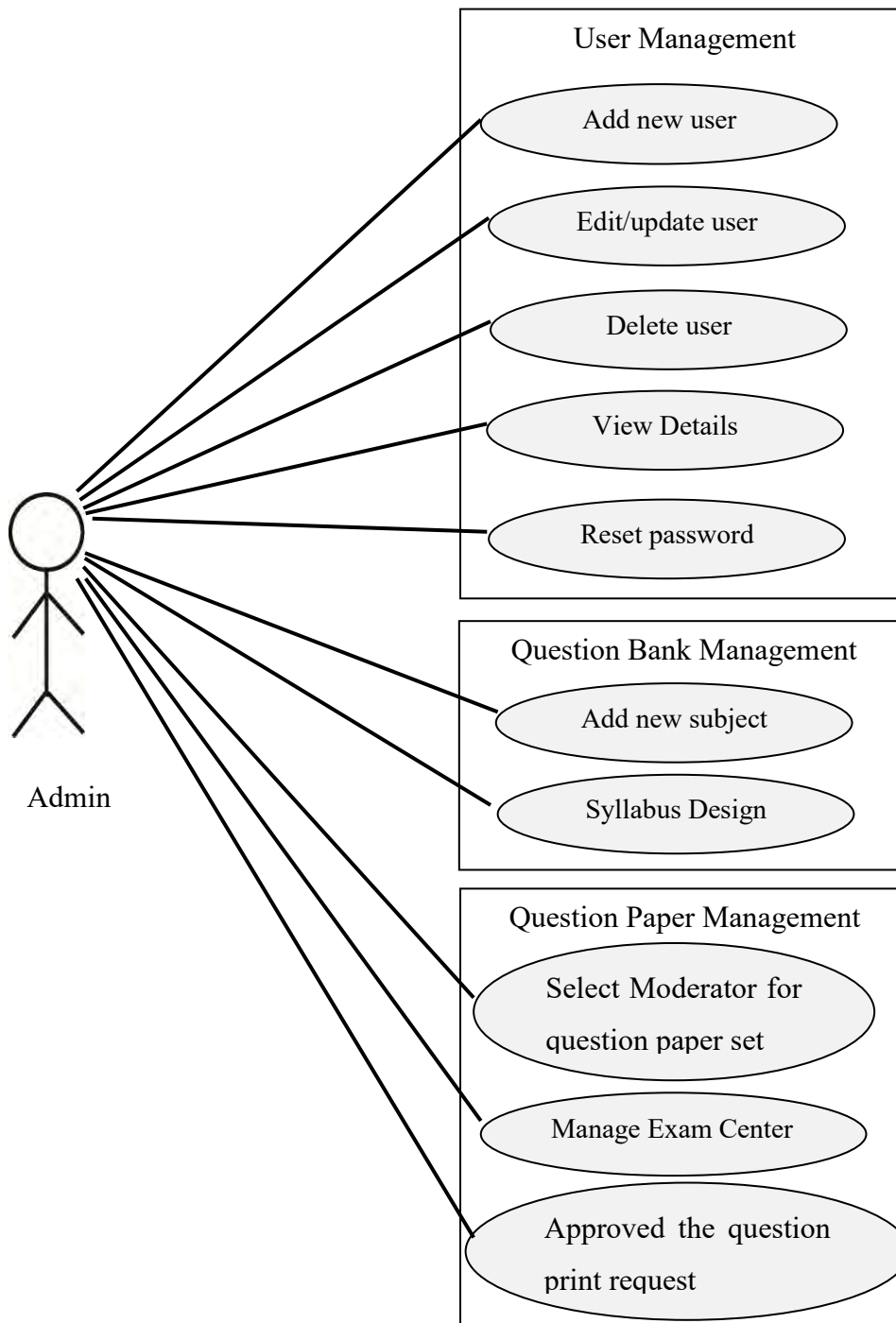


Figure 3.3: Use Case Diagram of Administrator.

3.2.1.2 Use Case Diagram of Moderator

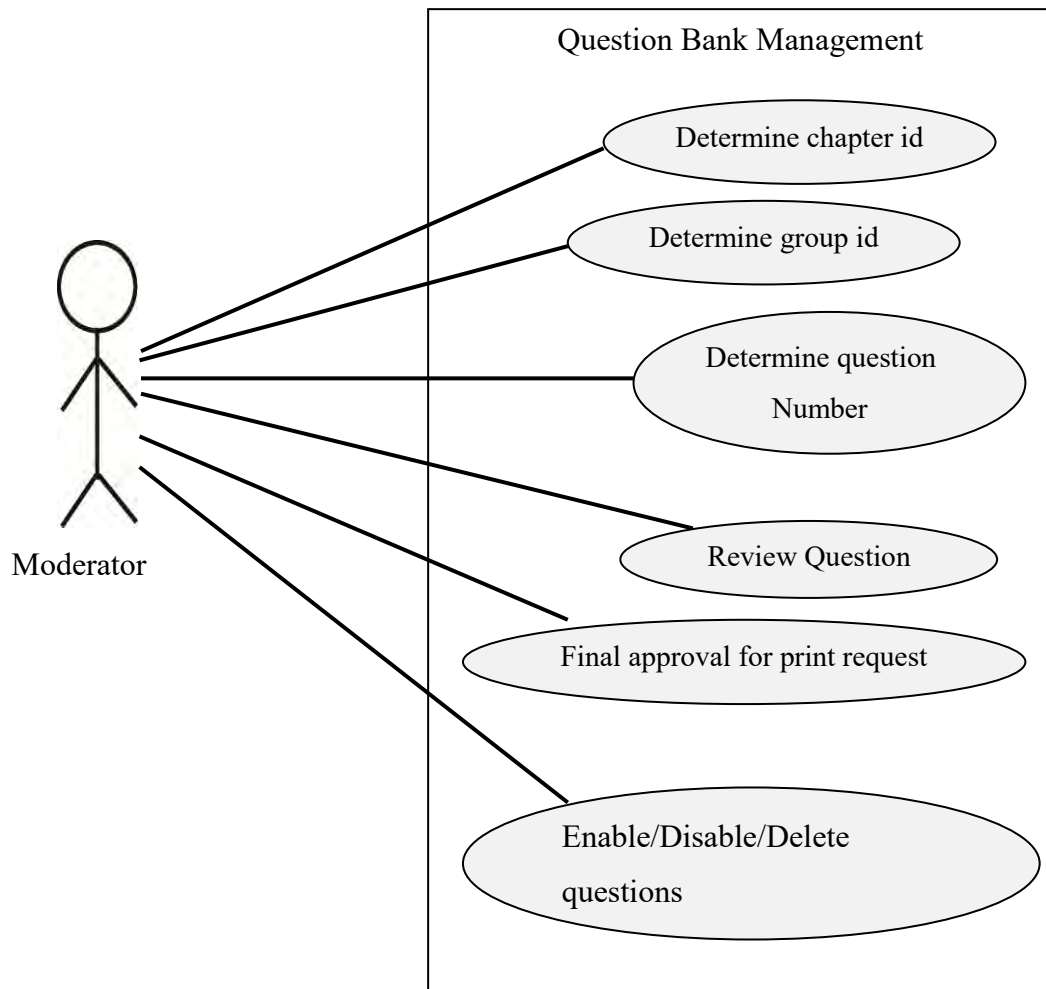


Figure 3.4: Shows Use Case Diagram of Moderator

3.2.1.3 Use Case Diagram of Composer

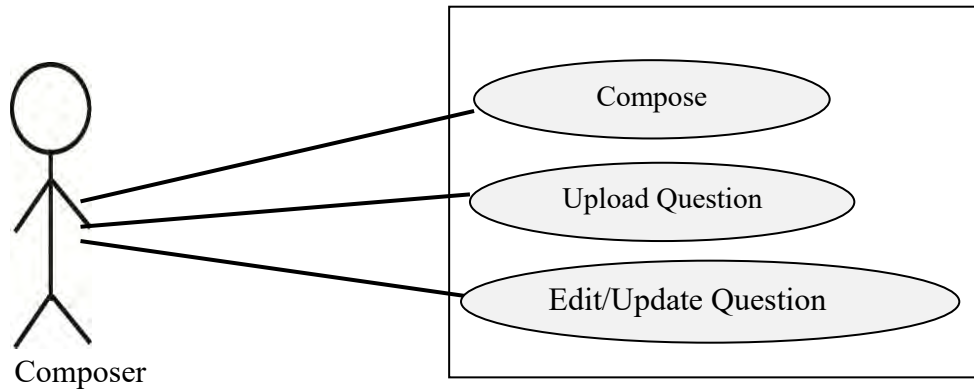


Figure 3.5: Shows Use case diagram of composer

3.2.1.4 Use Case Diagram of User

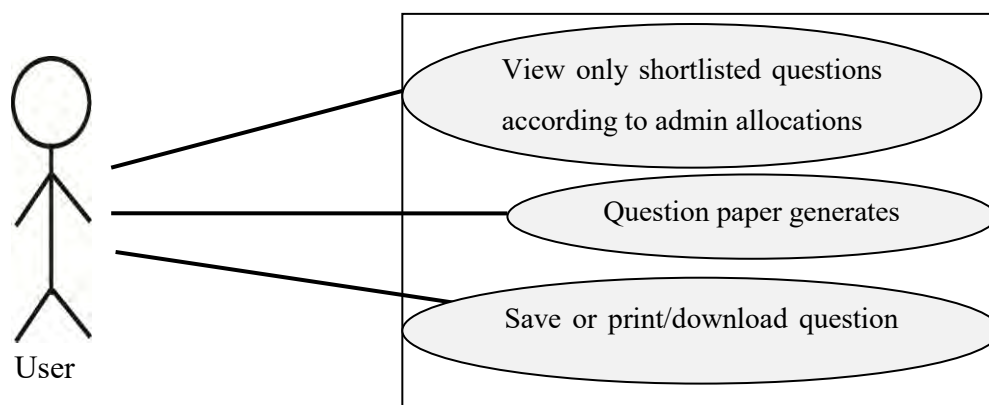


Figure 3.6: Shows Use case diagram of user.

3.2.2 Question Development Process

A graphical representation of the question development process is presented below in Figure 3.7

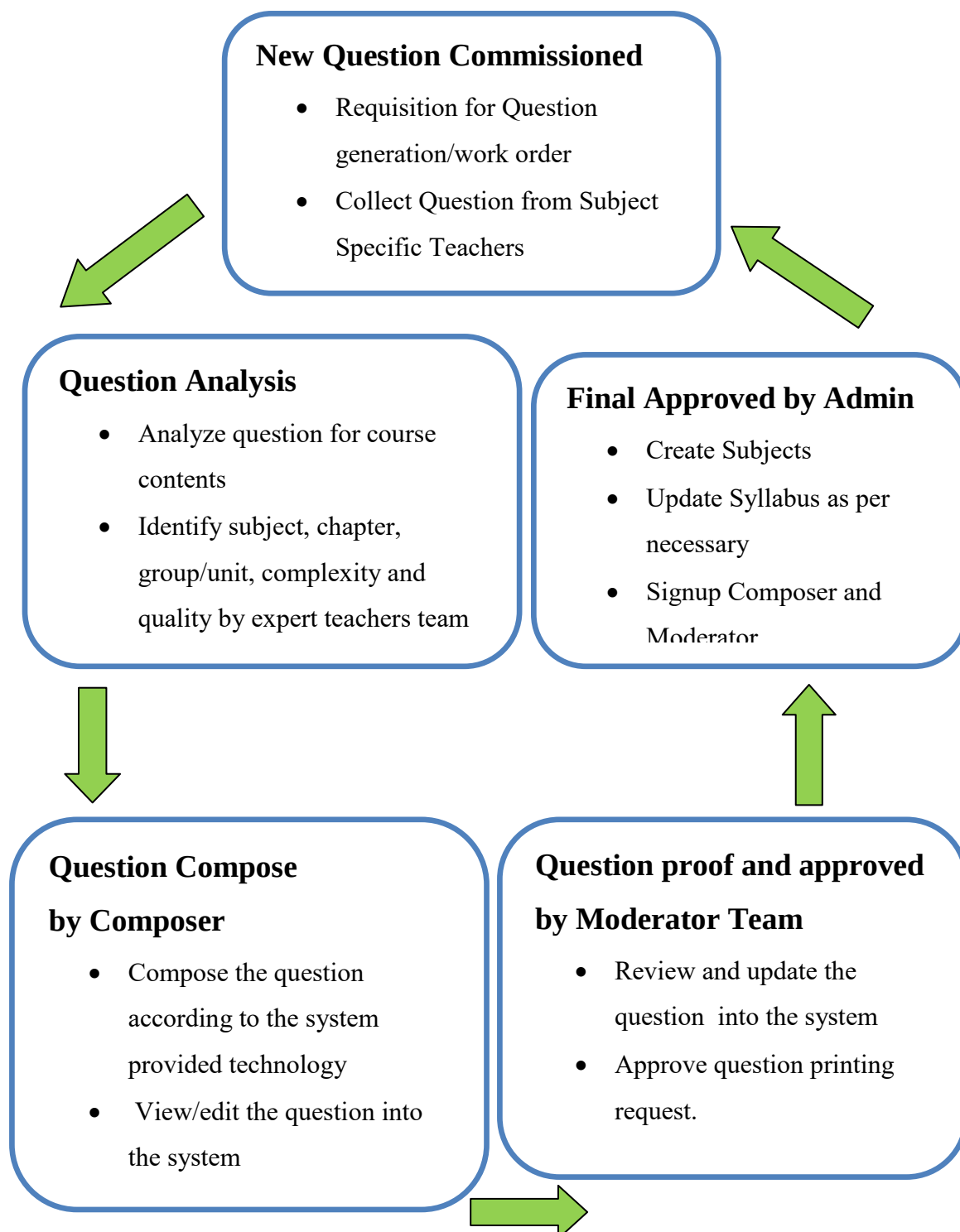


Figure 3.7: Life Cycle of Question Development Process

3.3 Database Design

A database is a collection of information, organized in such a way that a computer program can quickly select desired pieces of data. The computer program used to manage and query a database is known as a database management system (DBMS). Database design is the process of producing a detailed data model of a database. Databases are designed to offer an organized mechanism for storing, managing and retrieving information. This includes a detailed specification of data elements, data types, indexing options and other parameters residing in the DBMS data dictionary. Many models and languages are used for the design of the database. To design the database the Entity-Relationship Diagram (ERD) is used.

3.3.1 Entity and Attributes

For the entity relationship diagram, at first we need to identify an entity of this system. In this system, an important entity is question. Here, id is a primary key and composer_id, question_id, subject_id, chapter_no, group_no is a foreign key. General attributes are question_details, exercise, complexity, reference, status. Figure 3.8 shows Questions entity and attributes.

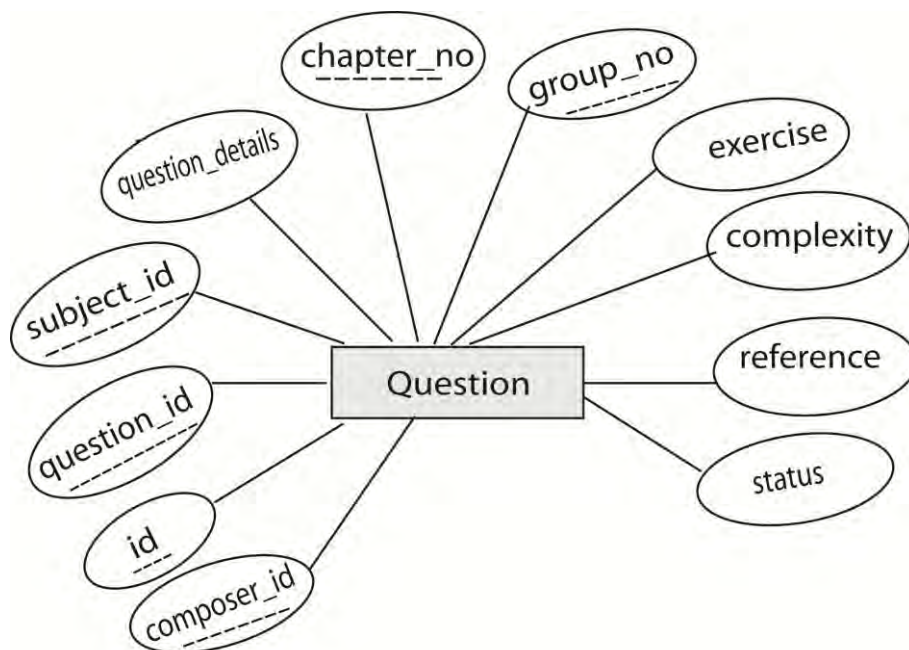


Figure 3.8: Questions Entity and Attributes

Another important entity is a chapter. Attributes as the primary key are id, admin_id, subject_id, chapter_no, group_no are a foreign key. General attributes are chapter name, status. Figure 3.9 shows chapter entity and attributes.

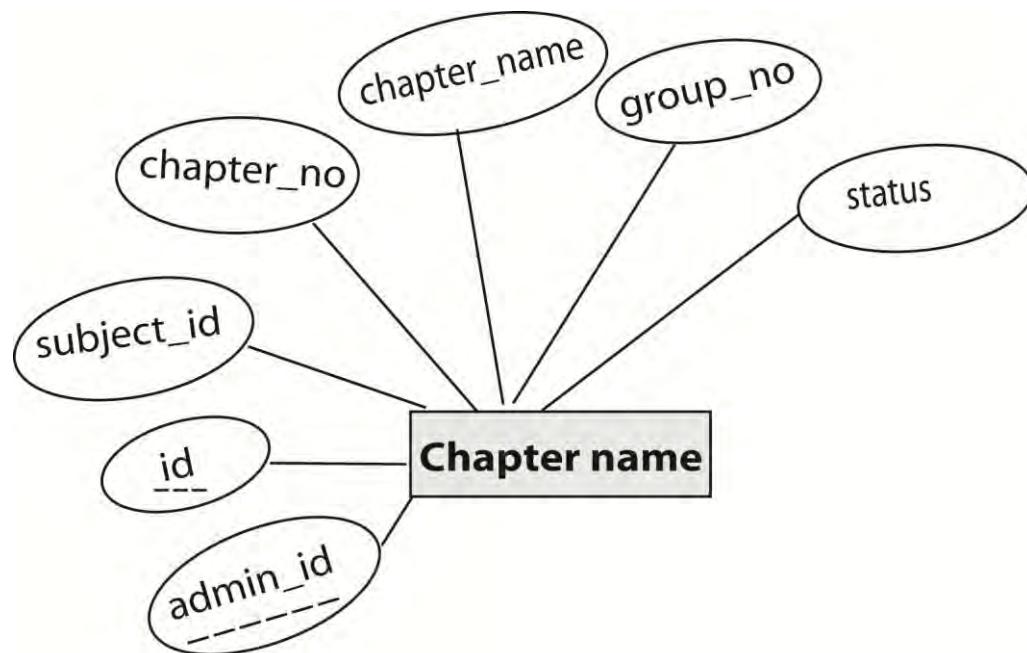


Figure 3.9: Chapter Entity and Attributes

Subject Assigned is another entity for permission to access subjects which is allocated by the administrator. Attributes as the primary key are id and user_id, subject_id, user_group are the foreign key. Status is a general key. Figure 3.10 shows subject assigned entity and attributes

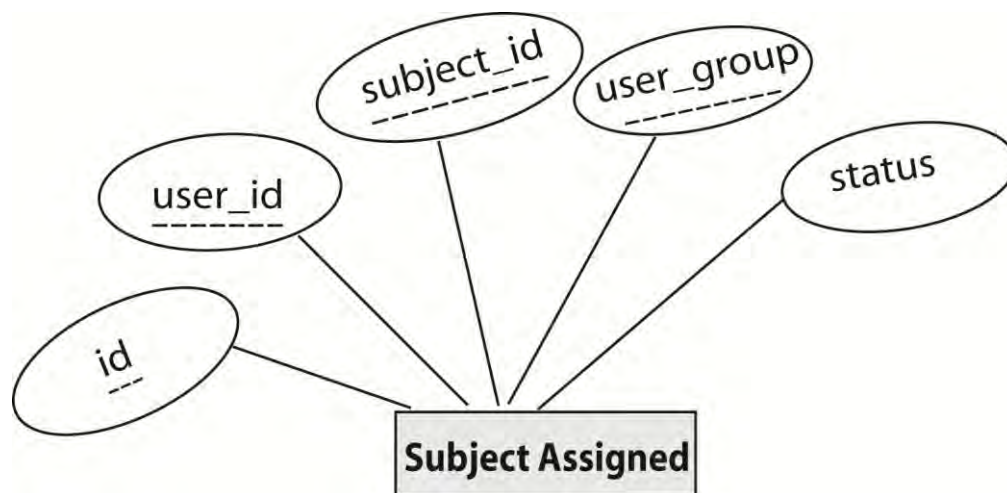


Figure 3.10: Subject Assigned Entity and Attributes

3.3.2 Entity Relationship Diagram (ERD)

An entity-relationship (E-R) diagram is a graphical representation of entities and their relationships to each other; typically used in organization of data within databases or information systems. There are three basic elements in the E-R diagram:

- Entities (tables) are the elements about which one seeks information. Boxes are commonly used to represent entities.
- Attributes are the data one collect about the entities. Ovals are used to represent attributes.
- Relationships provide the needed structure to draw information from multiple entities. Diamonds are normally used to represent relationships.

Figure 3.11 and 3.12 shows the E-R diagram of the proposed question paper generation system.

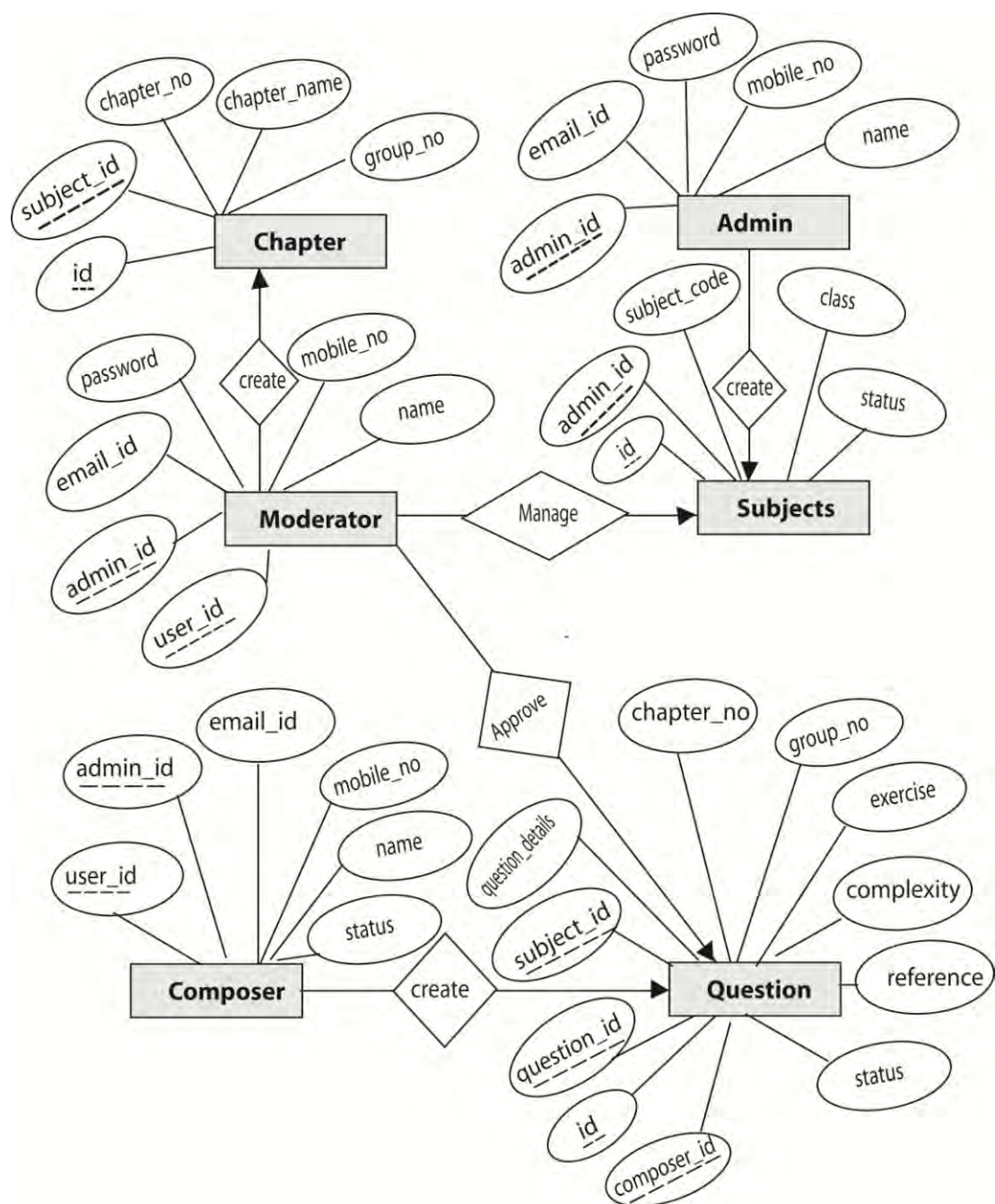


Figure 3.11: E-R Diagram for Question Bank

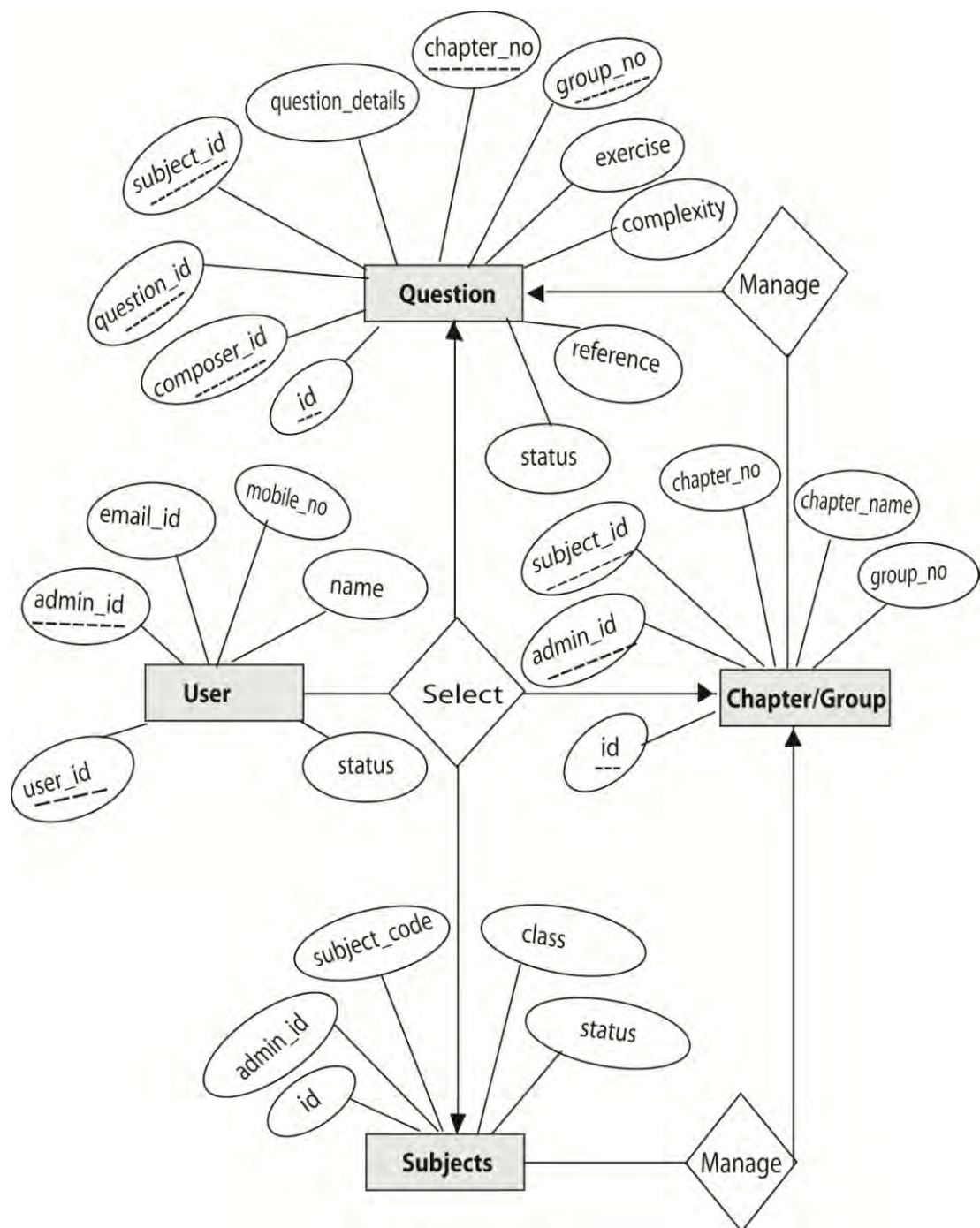


Figure 3.12: E-R Diagram for Question paper generation.

An entity-relationship diagram (ERD) is a data modeling techniques that can help define business processes and can be used as the foundation for a relational database. The three main cardinal relationships are One-to-one (1:1), One-to-many (1: M), Many-to-many (M: M). In this project, entities are administrators, composers, moderators, users, subject, chapter/group, and question as presented in Figure 3.11 & 3.12. Administrator's attributes are admin_id, email_id, password, name, mobile_no, etc. Moderator's attributes are moderator_id, email_id, password, name, mobile_no, etc. Composer's attributes are composer_id, email_id, password, name, mobile_no, etc. User's attributes are user_id, email_id, password, name, mobile_no, etc. Subject attributes are subject_id, subject_code, class, etc. Chapter attributes are subject_id, chapter_no, chapter_name, etc. Question attributes are subject_id, chapter_no, group_no, question_id, question_details, complexity, etc. An administrator can create many Subjects. So, administrator and subject relation is one-to-many (1: M). A composer can access one subject. So, composer and subject relation is one-to-one (1:1). Similarly, user and subject relation is many-to-many (M: M). For example a user selects a question for question paper generation, same time another user can also select the same question for question paper generation.

3.4 Implementation

3.4.1 Question Bank Management

After login, the system admin can create a user, composer, and moderator. An admin also can create a subject. Then, the admin can assign subjects to access the question bank. For user management and permission of access to specific subjects, the system maintain a table called *subjects_assigned_for_user*.

Table 3.1: Subject Assigned

id	user_id	subject_id	user_group	status
1	2	1	3	1

There are four types of user administrator, moderator, composer and user (Question paper generator). The user group is 1, the composer group is 2, moderator group is 3 and administrator group is 4. In table 3.1, user group 3 means user type is moderator.

Also id 1, subject_id =1 means the person's assigned subject is Physics. Here, status=1 means active. When a moderator login to the system, the specific subjects assigned to this person will be shown by the following query.

```
public function selectSubject()
{
    $user_id=$this->session->userdata('user_id');
    $this->db->select('*');
    $this->db->from('tbl_subject_name');
    $this->db->where('user_id',$user_id);
    $this->db->where('tbl_subject_name.status',1);
    $this->db->join('tbl_subjects_assigned_for_user',
        'tbl_subjects_assigned_for_user.subject_id = tbl_subject_name.id');
    $this->db->where('tbl_subjects_assigned_for_user.status',1);
    $this->db->where('tbl_subjects_assigned_for_user.user_group',3);
    $query_result= $this->db->get();
    $result=$query_result->result();
    return $result;
}
```

3.4.2 Question Compose

After login, the system composer can compose question according to the system provided technology and refer to moderator for review and approval. Then this new question will be stored in the database table.

Table 3.2: Creative_Question

id	compose r_id	subject_ id	questio n_id	question_details	Group_ no	exercise	complexi ty	status
1	3	2	20001		1	1,1.2,etc.	2	4

Here, composer_id 3 means 'Composer' who composes this question; subject_id 2 means subject name is 'Mathematics'; group_no 1 means 'Chapter' or 'Unit' no is 1; complexity 2 means 'Medium' type question and status 4 means this question is pending for moderator's approval.

After login, the system moderator can view details of the question and change status as active/inactive. This person also can edit or return to the composer for modification of the question if required.

Here, status=1 means this question is active for question paper generation. Others status means a user cannot access this question for question paper generation.

3.4.3 Question Paper Generation

The main feature of this module is to generate a question paper. After login the system, a user/moderator can select a subject for question paper generation. Then, the person can set the syllabus, number of questions and complexity of the question. Complexity and number of questions are maintained as shown in the following table

Table 3.3: Maintained Complexity in Question Paper Composition

user id	group no	qnum	easy	difficult	medium
1	1	4	1	1	2
1	2	2	1	1	0

Here, group no 1 means ‘Chapter’ or ‘Unit’ number is 3; qnum 4 means ‘the number of selected question’ is four.

The first row in Table 3.3 indicates that the selected number of questions is 4 where the number of ‘easy’ question is 1, the number of ‘difficult’ question is 1 and number of ‘medium’ question is 2, and the group or chapter no is 1. In this table, the 2nd row indicates that the selected number of question is 2 where the number of ‘easy’ question is 1, the number of ‘difficult’ question is 1 and number of ‘medium’ question is 0, and the group or chapter no is 2. These are all temporary data that are only created during question paper generation. When question paper generation is complete, this data will be removed from the database.

3.5 Coding

3.5.1 Model View Controller (MVC)

Standard programming is class and function-based. Here, CodeIgniter Framework is used which follows class and function base programming. With class and function, CodeIgniter decorates Model-View-Controller (MVC) which is another standard programming. Java J2EE, Asp.net also follows MVC based programming.

3.5.2 Error Handler

Error handler makes error handling easy. There are two types of error handler, 'HTTP 404 not found' and 'unauthorized access'. CodeIgniter Framework contains routes configuration under config folder and has parameter \$route ['404_override'] = 'errors/index' that just define HTTP 404 error calling the function index under error controller or class. Unauthorized access is managed by writing code for each function under each controller class which has been discussed in the next section.

3.5.3 Controller class

For the development of the project, some controllers such are Admin, Admin Login, Moderator, Moderator Login, Composer, Composer Login, User, User Login, Errors, and Basic were created. Admin controller is used to handling admin activity, and Admin Login controller used to manage admin login . Moderator controller is used to handling question moderation activity and Moderator Login controller used to manage Moderator login. Composer controller is used to handling composer activity and Composer Login controller used to manage Composer login. User controller is used to handling user activity, and User Login controller is used to manage user login. The general controller contains some functions that do not authorize. Error controller is used to handle 'HTTP 404 error'.

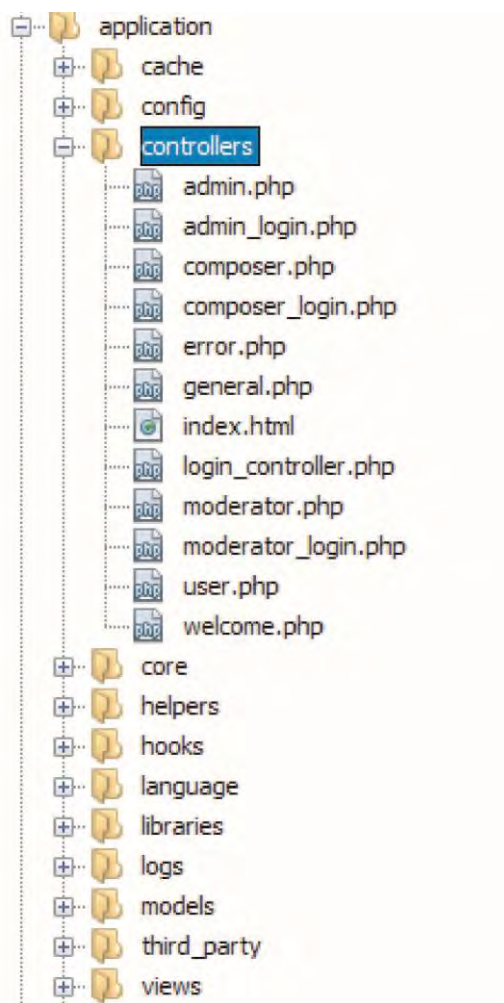


Figure 3.13: Controller Class

In order to ensure security, deferent controller is used in this project. For example, Admin Login controller is used to manage only admin login. After successful verification, Admin Login controller redirected to Admin controller where admin can handle her all activity; otherwise, admin cannot use Admin controller. For example, when admin calls a function like that 'admin/index' means admin is a class and index is a function and the following events occurre.

```
if(!$this->session->userdata('email_id')) redirect('admin_login');
```

Here, 'email_id' means Admin UserId and 'admin_login' mean different controller. This controller class index function views a login form in the web. Admin put userId and password in the form, and the following events occurred. The function name is checkLogin and query the following:

```

public function checkLogin() {
    $email = $this->input->post('email_id', true);
    $password = $this->input->post('password', true);
    $result=$this->admin_login_model->checkAdminLoginInfo($email, $password);
    if ($result) {
        $sdata = array();
        $sdata['name'] = $result->name;
        $sdata['admin_id'] = $result->admin_id;
        $sdata['email_id'] = $result->email_id;
        $sdata['login_status'] = true;
        $this->session->set_userdata($sdata);
        redirect('admin');
    } else {
        $sdata = array();
        $sdata['message'] = 'Email Id Or Password Invalide.';
        $this->session->set_userdata($sdata);
        redirect("admin_login/index");
    }
}

```

```

public function checkAdminLoginInfo($email, $password) {
    $this->db->select('*');
    $this->db->from('tbl_super_admin');
    $this->db->where('email_id', $email);
    $this->db->where('password', md5($password));
    $this->db->where('activation_status', 1);
    $query_result = $this->db->get();
    $result = $query_result->row();
    return $result;
}

```

After successful verification, the user is redirected to admin index page.

3.5.4 Model class

For database programming, some model class were created such as Admin Model, Admin Login Model, Moderator Model, Moderator Login Model, Composer Model, Composer Login Model, User Model, User Login Model.

Admin Login Model is used to process admin login details; Admin Model is used to process admin information. Moderator Login Model is used to process moderator login details, Moderator Model is used to process moderator information. Composer Login Model is used to process composer login details, Composer Model is used to process Composer information. User Login Model is used to process user login details, User Model is used to process user information.

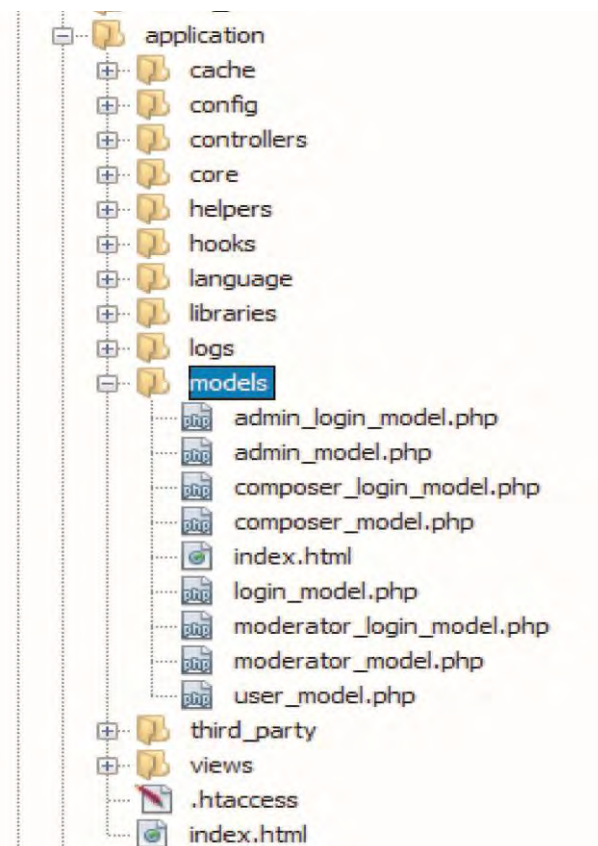


Figure 3.14: Model Class

3.5.5 Database Table

There are 22 tables used to complete the project. The following is the list of tables.

Table	Action	Rows	Type	Collation	Size	Overhead
tbl_chapter_name	Browse Structure Search Insert Empty Drop	166	InnoDB	latin1_swedish_ci	48.0 KiB	-
tbl_class_name	Browse Structure Search Insert Empty Drop	2	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_composer	Browse Structure Search Insert Empty Drop	3	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_creative_question	Browse Structure Search Insert Empty Drop	34	InnoDB	latin1_swedish_ci	80.0 KiB	-
tbl_exam_name	Browse Structure Search Insert Empty Drop	5	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_exercise	Browse Structure Search Insert Empty Drop	43	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_general_question	Browse Structure Search Insert Empty Drop	2	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_group	Browse Structure Search Insert Empty Drop	50	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_img_up_history	Browse Structure Search Insert Empty Drop	25	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_institute_name	Browse Structure Search Insert Empty Drop	2	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_instruction	Browse Structure Search Insert Empty Drop	1	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_mcq_question	Browse Structure Search Insert Empty Drop	6	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_moderator	Browse Structure Search Insert Empty Drop	3	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_qtype	Browse Structure Search Insert Empty Drop	3	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_subjects_assigned_for_user	Browse Structure Search Insert Empty Drop	10	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_subject_name	Browse Structure Search Insert Empty Drop	11	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_super_admin	Browse Structure Search Insert Empty Drop	3	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_user	Browse Structure Search Insert Empty Drop	5	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_user_login	Browse Structure Search Insert Empty Drop	3	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_user_selected_question	Browse Structure Search Insert Empty Drop	0	InnoDB	latin1_swedish_ci	16.0 KiB	-
tbl_user_selected_unit	Browse Structure Search Insert Empty Drop	5	InnoDB	latin1_swedish_ci	16.0 KiB	-
22 tables	Sum	382	InnoDB	latin1_swedish_ci	~448.0 KiB	0 B

Figure 3.15: Screenshot of the list of database tables

- The table subject_name is used for containing subject information which is accessed by the administrator.
- The table chapter_name is used for containing chapter information such as chapter name, chapter no for each subject.
- The table group_name is used for containing group or unit information such as group name, group no for each subject.
- The exercise table is used for containing exercise no of each chapter.
- The table creative_question is used for containing all creative question information such as question id, subject id, group no, exercise, question description, complexity of each subject and as similar table. Similarly mcq_question table is used for containing all MCQ information.
- The table class_name is used for containing class name of each subject which is accessed by the administrator.
- The table super_admin is used for administrator information.

- The table moderator is used for moderator information. The table composer is used for composer information.
- The table user_login is used for user information.
- The table user_selected_unit is used for user selected chapter or group information.
- The table user_selected_question is used for user selected question.

CHAPTER 4

EXPERIMENTAL RESULT

4.1 Introduction

The proposed Question Paper Generation System has different essential features. Screen shots of some of the main features and their explanation are given in the next sections.

4.2 Questions Bank Module

This project has two main features such as Development a web-based question bank and automated question paper generation. The Question Bank Module is explained below.

4.2.1 Admin Sections

Figure 4.1 shows the graphical user interface (GUI) view. This page is displayed after log in to the system using Admin credential.



Figure 4.1: A Screenshot of the Admin Home Page

This page contains four main menus. First one user management, second one composer management, Third one moderator management and forth one is subjects management. The main feature of user management is user registration, edit/update user information, reset the password, subjects assigned, etc. All users of the website

will display in the page. From this page, admin can create/edit/delete users. Admin can also make the status of users active/inactive. If a user is inactive, then this person will be in the user list but will not be able to log-in. Composer and Moderator Management menus are same as User Management. The main feature of Subject management menu is to create/update/manage all subject information. Figure 4.2, 4.3, and 4.4 shows the main features of relevant admin sections.

AUTOMATED QUESTION PAPER GENERATION SYSTEM - ADMIN

Home User Management Composer Management Moderator Management Subject

USER REGISTRATION:

Contact Mobile: 01911

Email ID:

Password:

Name of User:

Address:

Submit Save

Figure 4.2: A Screenshot of the User Registration Form

AUTOMATED QUESTION PAPER GENERATION SYSTEM - ADMIN

Home User Management Composer Management Moderator Management Subject

User List User Registration

User Name	Mobile Number	E-mail/User ID	Status	Action
Monir Ahamed	01911923349	user@gmail.com	Active	Assigned New subject View Subject Edit Reset Password Delete
Atkur Rahman	01719974733	user2@gmail.com	Active	Assigned New subject View Subject Edit Reset Password Delete
Milu Ahamed	01911923349	user4@gmail.com	Active	Assigned New subject View Subject Edit Reset Password Delete

powered by 0.16.36 scsscss.

Figure 4.3: A Screenshot of the View All User Page

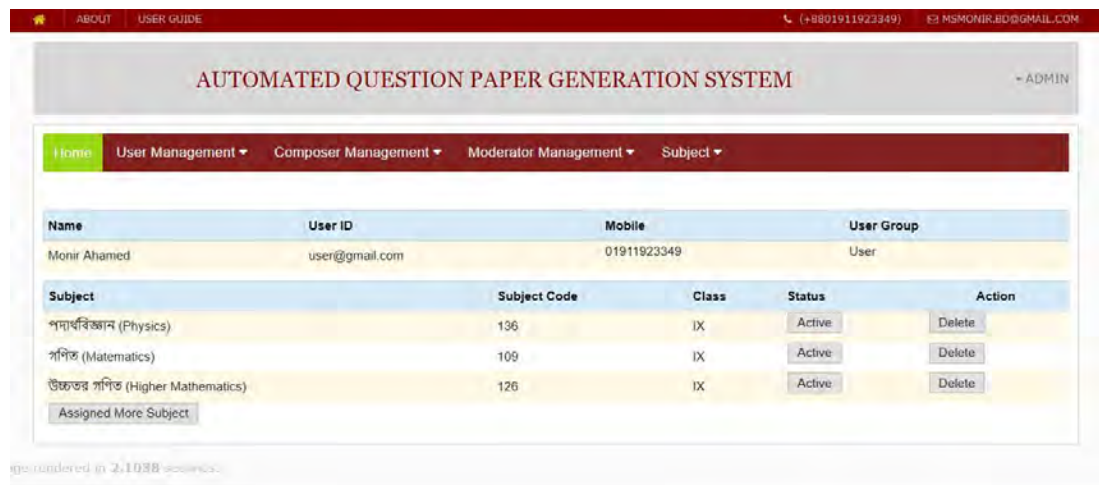


Figure 4.4: A Screenshot of the View Subject Assigned Page

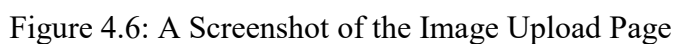
4.2.2 Composer Sections

After sign in, a composer will automatically move to this page. Figure 4.5 shows the view of the composer home page.



Figure 4.5: A Screenshot of the Composer Home Page

This page contains four menus. The first one is the compose menu. This feature is to compose a new question and upload image. The second one is upload pending menu, which shows all questions previously composed but not yet uploaded. Here, the composer can edit/update this question and upload question for the approval of the moderator. The third one is approval pending menu, which shows all questions previously uploaded but not yet approved. Here, composer can edit/update and upload question again. The fourth one is return question menu which shows all questions returned from the moderator to the composer for change or correction.



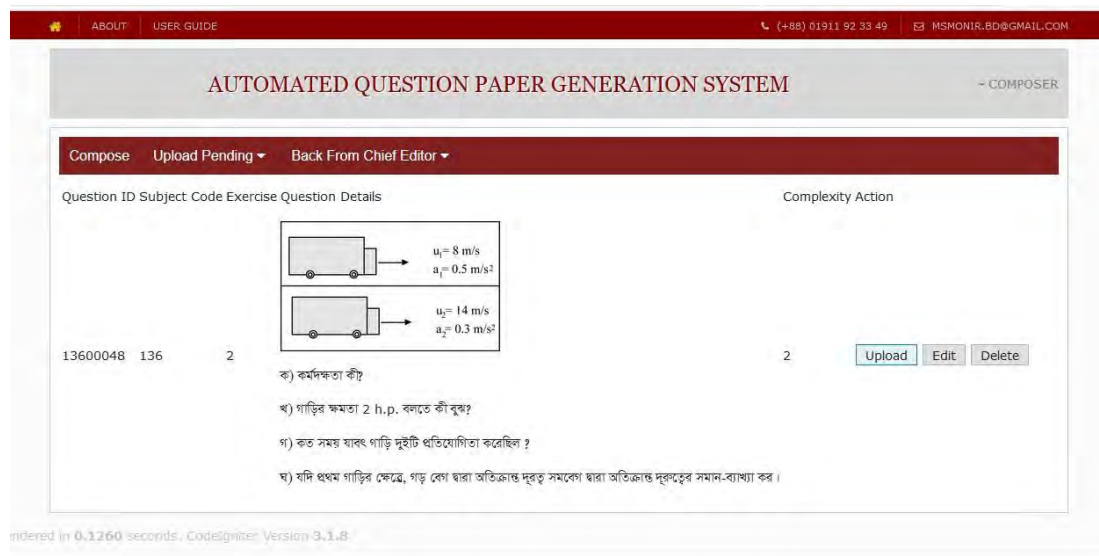


Figure 4.8: A Screenshot of the Composer View Question Page

4.2.3 Moderator Sections

After sign in, a moderator will automatically move to this page. Figure 4.9 shows the home page view of the moderator.

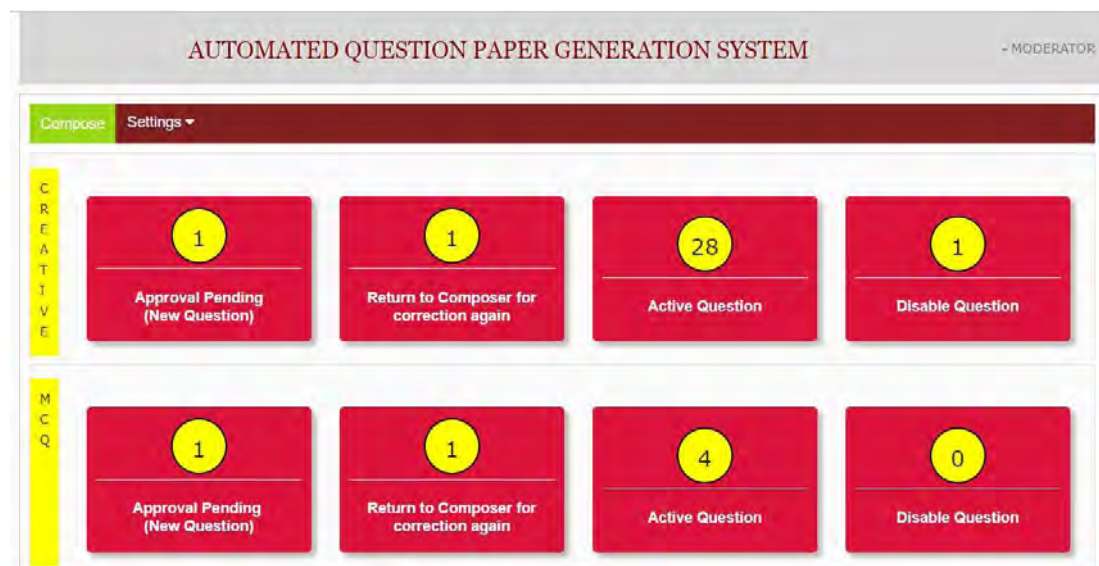


Figure 4.9: A Screenshot of the Moderator Home Page

This page contains two menus. First one is question management and second one is chapter/unit management. The purpose of the question management menu is

questions approval, edit, status (active/disable) etc. The purpose of the chapter/unit management menu is to create/update/manage chapter/unit for each subject. A moderator can set the default number of the question for each subject. Figure 4.10, 4.11, 4.12, 4.13, and 4.14 shows the screenshots of the relevant pages.

The screenshot displays the 'AUTOMATED QUESTION PAPER GENERATION SYSTEM' interface. At the top, there is a navigation bar with links for 'ABOUT' and 'USER GUIDE', along with contact information: '(+88) 01911 92 33 49' and 'MSMONIR.BD@GMAIL.COM'. The main header area includes the system name and a 'MODERATOR' role indicator.

The central part of the interface features a table with the following columns: Question ID, Subject Code, Exercise, Question Details, Complexity, and Action. The table contains one row with the following data:

Question ID	Subject Code	Exercise	Question Details	Complexity	Action
13600048	136	2	ক) কর্মদিবসতা কী? খ) গাড়ির ক্ষমতা 2 h.p. বলতে কী বুঝ? গ) কত সময় যাবে গাড়ি দুইটি প্রতিযোগিতা করেছিল? ঘ) যদি প্রথম গাড়ির ক্ষেত্রে, গাড়ি বেগ দ্বারা অতিক্রান্ত দূরত্ব সমবেগ দ্বারা অতিক্রান্ত দূরত্বের সমান-ব্যাখ্যা কর।	2	Approved Return Disable Edit

At the bottom of the page, it states: 'Page rendered in 0.1273 seconds. CodeIgniter Version 3.1.8'.

Figure 4.10: A Screenshot of the Approved Page

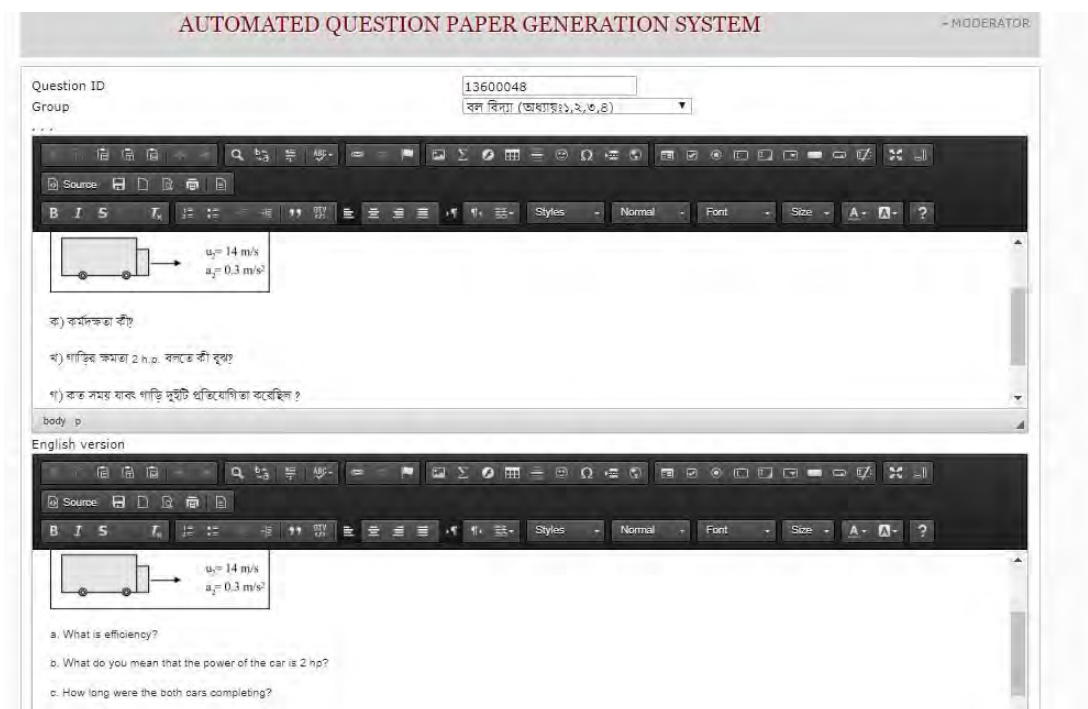


Figure 4.11: A Screenshot of the Moderator Edit Page

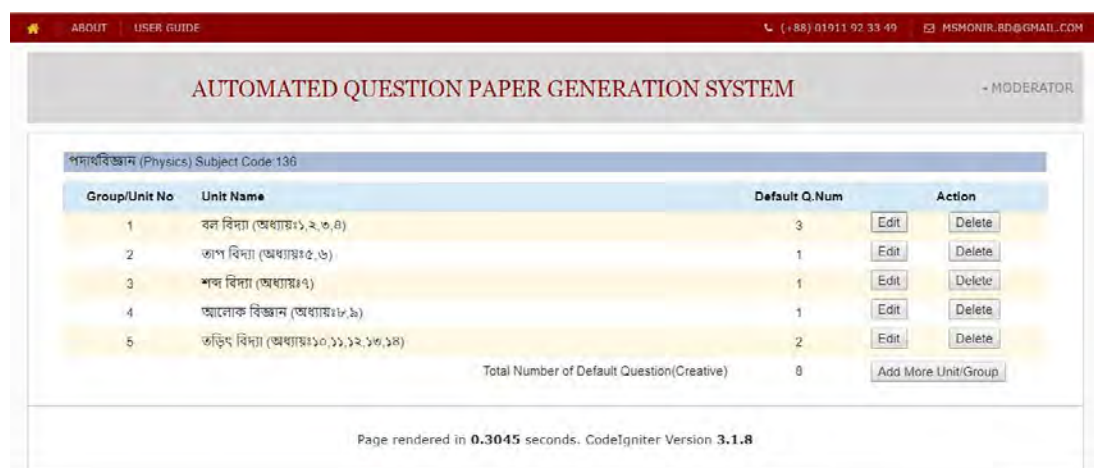


Figure 4.12: A Screenshot of the Syllabus Settings Page

পদার্থবিজ্ঞান (Physics) Subject Code:136				
Chapter No	Chapter Name	Default Q.Num(MCQ)	Action	
1	ভৌত রাশি ও পরিমাপ	2	Edit	Delete
2	গতি	2	Edit	Delete
3	বল	2	Edit	Delete
4	কাজ, ক্ষমতা ও শক্তি	2	Edit	Delete
5	পদার্থের অবস্থা ও চাপ	2	Edit	Delete
6	বস্তুর উপর চাপের প্রভাব	1	Edit	Delete
7	তরঙ্গ	2	Edit	Delete
8	আলোর প্রতিফলন	2	Edit	Delete
9	আলোর প্রতিসরণ	2	Edit	Delete
10	স্থির তড়িৎ	2	Edit	Delete
11	চল তড়িৎ	2	Edit	Delete
12	তড়িৎের চৌম্বক ক্রিয়া	2	Edit	Delete
13	আধুনিক পদার্থবিজ্ঞান ও ইলেক্ট্রনিক্স	1	Edit	Delete
14	জীবন বাঁচাতে পদার্থবিজ্ঞান	1	Edit	Delete
Total Number of Default Question		25	Add More Chapter	

Figure 4.13 A Screenshot of the Chapter Settings Page

[ABOUT](#)
[USER GUIDE](#)
(+88) 01911 92 33 49
MSHOWIR.BD@GMAIL.COM

AUTOMATED QUESTION PAPER GENERATION SYSTEM MODERATOR

পদার্থবিজ্ঞান (Physics) Subject Code:136

Chapter No	Chapter Name	Default Q.Num(MCQ)	Action
1	ভৌত রাশি ও পরিমাপ	2	Update Cancel

Page rendered in 0.1371 seconds. CodeIgniter Version 3.1.8

Figure 4.14 A Screenshot of the Chapter Edit Page

4.3 Questions Paper Generation Module

Automated Question Paper Generation is the most important module of this project. This system includes two different types of question paper generation. One is for public examination and another is for other examinations.

4.3.1 Public Examination Module

Automated Question Paper Generation for Public Examination (JSC, SSC, HSC etc) is explained below.

4.3.1.1 Question Pattern Set

For public examination (JSC, SSC, HSC etc.), a registered moderator only can set the pattern, i.e., total number of questions, chapter and complexity level of each question. To ensure the security, moderator remains unaware of the generated question.

Figure 4.15, 4.16, 4.17, 4.18, and 4.19 shows screenshots of the question pattern.

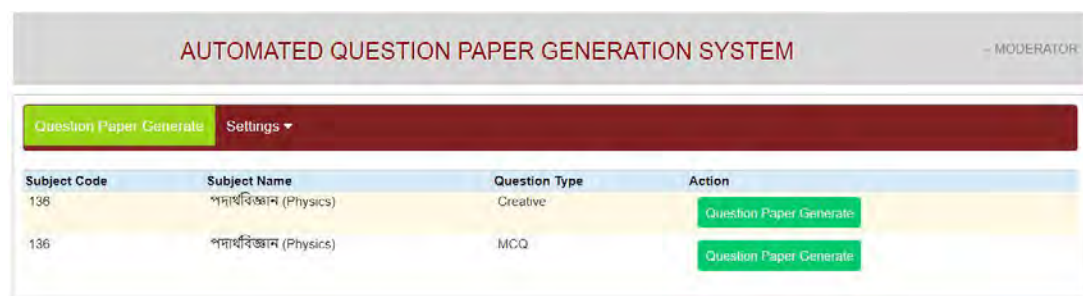


Figure 4.15 A Screenshot of the Subject Selection Page

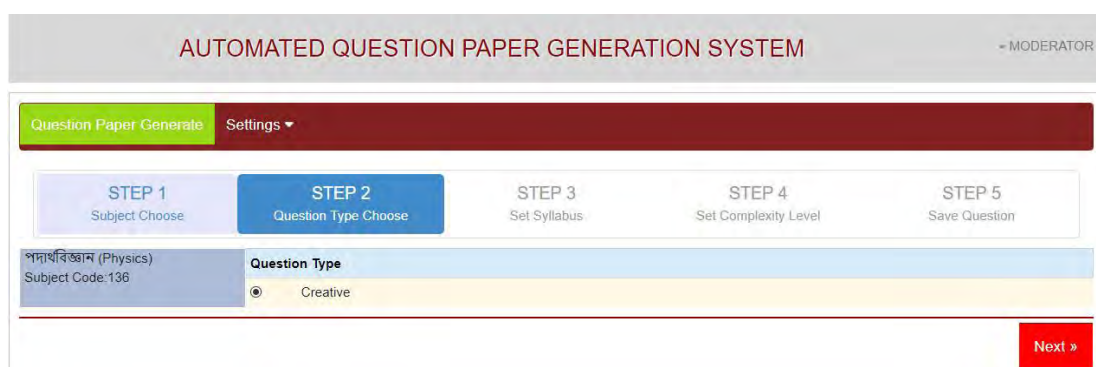


Figure 4.16: A Screenshot of the Question Type Selection Page

STEP 1 Subject Choose		STEP 2 Question Type Choose		STEP 3 Set Syllabus		STEP 4 Set Complexity Level		STEP 5 Save Question	
Group/Unit No	Unit Name	Default Q.Num	Number of Ques	Action					
1	বন বিদ্যা (অধ্যায়ঃ১,২,৩,৪)	3	3	Update	Delete				
2	তাপ বিদ্যা (অধ্যায়ঃ৫,৬)	1	1	Update	Delete				
3	শব্দ বিদ্যা (অধ্যায়ঃ৭)	1	1	Update	Delete				
4	আলোক বিজ্ঞান (অধ্যায়ঃ৮,৯)	1	1	Update	Delete				
5	তড়িৎ বিদ্যা (অধ্যায়ঃ১০,১১,১২,১৩,১৪)	2	2	Update	Delete				
Total		8	8						

Figure 4.17 A screenshot of the set syllabus and number of questions each unit page

STEP 1 Subject Choose		STEP 2 Question Type Choose		STEP 3 Set Syllabus		STEP 4 Set Complexity Level		STEP 5 Save Question	
Complexity Level Easy Difficult Medium 50% 25% 25%									

Figure 4.18 A Screenshot of the Complexity Set Page

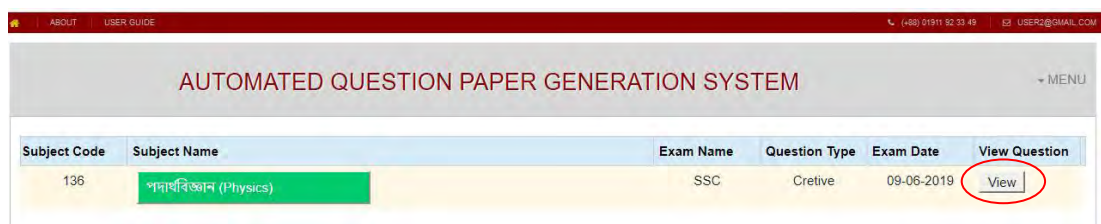
AUTOMATED QUESTION PAPER GENERATION SYSTEM		- MODERATOR	
Question Paper Generate Settings			
Question Summary			
		Total Question	8
		Easy Question	4
		Difficult Question	2
		Medium Question	2

Figure 4.19 A Screenshot of the Question Summary View Page

4.3.1.2 Question Paper Generation at Exam Centers

This module is created for security purpose. So, only authorized exam center can print question paper after a stipulated time. Whenever the authorized exam center request to generate a question paper, the system generates a random one time password (OTP) send to a specified cell phone or email address. The requesters then use this OTP to download or print the question paper from the secured server.

Figure 4.20, 4.21, 4.22 and 4.23 shows the screenshots of these sections.



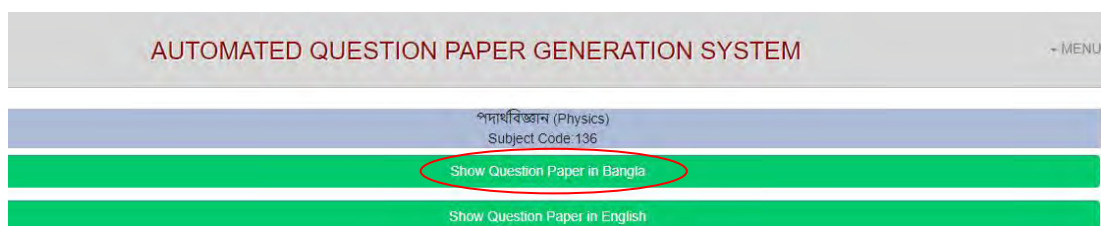
The screenshot shows the 'AUTOMATED QUESTION PAPER GENERATION SYSTEM' interface. At the top, there is a navigation bar with 'ABOUT' and 'USER GUIDE' links, and contact information: '(+88) 21911 52 33 45' and 'USER2@GMAIL.COM'. Below the header, there is a table with columns: 'Subject Code', 'Subject Name', 'Exam Name', 'Question Type', 'Exam Date', and 'View Question'. The table contains one row with the following data: Subject Code '136', Subject Name 'পদার্থবিজ্ঞান (Physics)' (highlighted in green), Exam Name 'SSC', Question Type 'Cretive', Exam Date '09-06-2019', and a 'View Question' button (highlighted with a red circle). A '+ MENU' link is visible in the top right corner.

Figure 4.20 A Screenshot of the Subject Selection Page



The screenshot shows the 'AUTOMATED QUESTION PAPER GENERATION SYSTEM' interface for security verification. It features two input fields: 'Security Key1 (PIN CODE)' with the value '136089' and 'Security Key2 (OTP)' with the value '346712'. Below these fields is a 'Submit' button and a 'Verify Key' button (highlighted with a red circle). A '+ MENU' link is visible in the top right corner.

Figure 4.21 A Screenshot of the Security Key Verify Page



The screenshot shows the 'AUTOMATED QUESTION PAPER GENERATION SYSTEM' interface for language selection. It displays the subject 'পদার্থবিজ্ঞান (Physics)' and 'Subject Code: 136'. Below this, there are two buttons: 'Show Question Paper in Bangla' (highlighted with a red circle) and 'Show Question Paper in English'.

Figure 4.22 A Screenshot of the Question Paper Language Selection Page

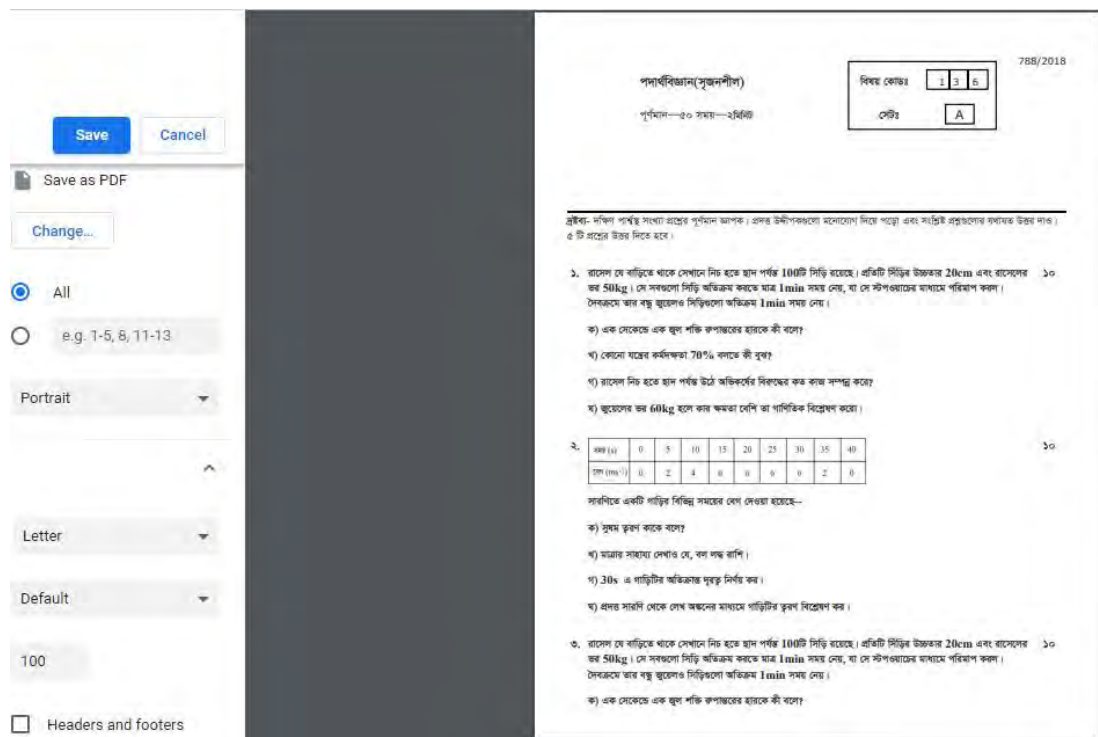


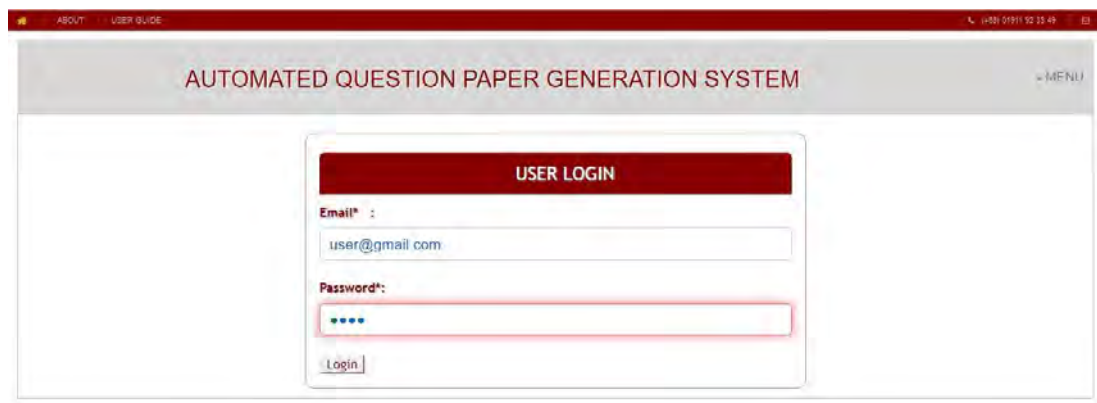
Figure 4.23 A Screenshot of the Question Paper Print or Save as PDF Page

4.3.2 Other Examination Module

For other examinations like a class test, semester final and yearly test, the question paper generation process is explained below.

4.3.2.1 User Login

Only authorized users will have access to this question paper generation. A user can log in to the system to gain access and then log out when the access is no longer needed. Figure 4.24 shows the screenshot of the user login page.

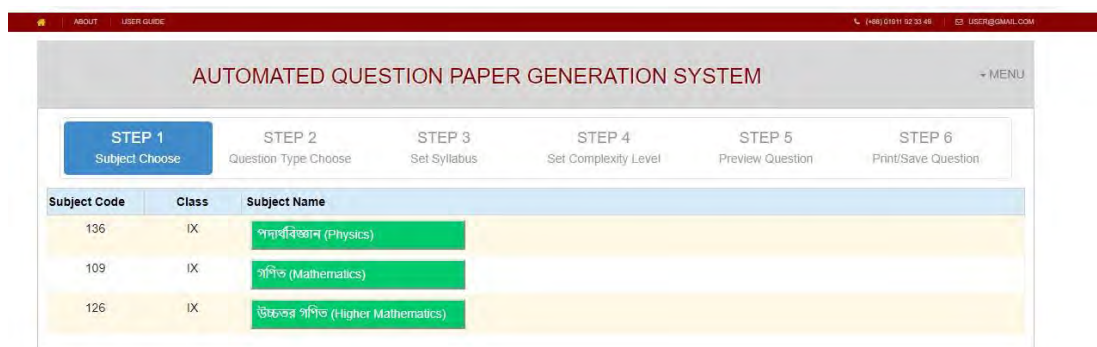


The screenshot shows the 'AUTOMATED QUESTION PAPER GENERATION SYSTEM' login interface. It features a 'USER LOGIN' box with fields for 'Email*' (containing 'user@gmail.com') and 'Password*' (masked with dots). A 'Login' button is at the bottom of the box. The page has a red header with 'ABOUT' and 'USER GUIDE' links, and a 'MENU' link in the top right.

Figure 4.24 A Screenshot of the User Login Page

4.3.2.2 Subject Selection

After sign in, a user will automatically direct to this page. Figure 4.25 shows the screenshot of the user home page.



The screenshot shows the 'AUTOMATED QUESTION PAPER GENERATION SYSTEM' home page. It features a progress bar with six steps: STEP 1 (Subject Choose), STEP 2 (Question Type Choose), STEP 3 (Set Syllabus), STEP 4 (Set Complexity Level), STEP 5 (Preview Question), and STEP 6 (Print/Save Question). Below the progress bar is a table listing subjects.

Subject Code	Class	Subject Name
136	IX	পদার্থবিজ্ঞান (Physics)
109	IX	গণিত (Mathematics)
126	IX	উচ্চতর গণিত (Higher Mathematics)

Figure 4.25 A Screenshot of the User Home Page

In this page, a user can see subjects list which is allocated by admin. For a question paper generation, the user can select a subject and then the following page (Figure 4.26) will be loaded in the user interface.

4.3.2.3 Question Type Selection

In this page, a user can select the type of question using radio button; then click on the next button. Figure 4.26 shows the screenshot of question type selection page.

AUTOMATED QUESTION PAPER GENERATION SYSTEM

STEP 1: Subject Choose | **STEP 2: Question Type Choose** | STEP 3: Set Syllabus | STEP 4: Set Complexity Level | STEP 5: Preview Question | STEP 6: Print/Save Question

পদার্থবিজ্ঞান (Physics)
Subject Code: 138

Question Type

☒ Creative
☐ MCQ
☐ General

Next »

Figure 4.26 A Screenshot of the Question Type Selection Page

4.3.2.4 Syllabus and Number of Question Set

This page automatically displays the standard format of syllabus and the number of questions for each chapter/unit. A user can change this syllabus format if necessary, and then click on the next button to proceed. Figure 4.27 shows the screenshot of the question type selection page.

AUTOMATED QUESTION PAPER GENERATION SYSTEM

STEP 1: Subject Choose | STEP 2: Question Type Choose | **STEP 3: Set Syllabus** | STEP 4: Set Complexity Level | STEP 5: Preview Question | STEP 6: Print/Save Question

Group/Unit No	Unit Name	Default Q.Num	Number of Ques	Action
1	বল বিদ্যা (অধ্যায়ঃ১,২,৩,৪)	3	3	Update Delete
2	তাপ বিদ্যা (অধ্যায়ঃ৫,৬)	1	1	Update Delete
3	শব্দ বিদ্যা (অধ্যায়ঃ৭)	1	1	Update Delete
4	আলোক বিজ্ঞান (অধ্যায়ঃ৮,৯)	1	1	Update Delete
5	তড়িৎ বিদ্যা (অধ্যায়ঃ১০,১১,১২,১৩,১৪)	2	2	Update Delete
Total		8	8	

Next »

Figure 4.27 A Screenshot of the set syllabus and number of questions each unit page

4.3.2.5 Set Complexity Level Page

This page automatically displays the standard format of the complexity level. A user can change the complexity level format and then click on next button to proceed. Figure 4.28 shows the screenshot of the complexity level page.

Figure 4.28 A Screenshot of the Complexity set page

4.3.2.6 Preview Questions Page

This page automatically displays the questions which were randomly selected as per user's choice of syllabus and complexity. A user can change, remove or add more question and then click on the next button to proceed. Figure 4.29 shows the screenshot of the question type selection page.

Figure 4.29 A Screenshot of the Preview Questions for Change, Remove or Add More Questions

4.3.2.7 Set Timetable

Here, a user can set the question paper headers like the name of exam, full marks, and duration of exam, set code, etc.; then click on 'Show Question' in Bengali button

for Bangla version or click on ‘Show Question in English’ button for English version. Figure 4.30 shows the question paper timetable.

Figure 4.30 A Screenshot of the Set Timetable Page

4.3.2.8 Question Paper Save or Print

The last step of this question paper generation process is the question paper print or saves. In this step, a user will view the following page.

Figure 4.31 A Screenshot of the Question Paper Print View Page

CHAPTER 5

CONCLUSION

5.1 Conclusion

The application of Question Bank creation and Question Paper Generation are very helpful for educational institutions. The automated web-based question paper generation software is expected to save the time of the instructor in comparison with the manual question paper generation. Day by day, technology is growing with its own potential. There are no standardized methods to ensure the quality of question paper. This system helps to generate customized question paper with the desired format which will be based on the instructor's requirement in less time. Our system is robust to question paper mishandling and replaces the manual operation of the traditional method.

The implemented work narrates an automated system that heads away from the traditional process of paper generation to an automated process, by giving controlled entry to the resources that is attained by involving users and their roles in the school and colleges. We have also considered the significance of randomization in the process of question paper generation. Our system has used a logical approach to randomizing the nature of question selection and avoids duplication of questions. Hence, we conclude that our system aims at generating question papers by fulfilling the basic constraints such as difficulty level, paper format, and coverage of the syllabus and limited possibility of paper leak and threats.

5.2 Future Work

In the current research, this software has been made to accommodate the generation of question paper in some subjects and some examination i.e., JSC, SSC, and HSC. In the near future, we will apply such operations for all subjects and in all other examination.

This version does not contain HTTPS due to the cost associated with an SSL certificate. But such implementation can be easily incorporated into the software in our extended version.

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